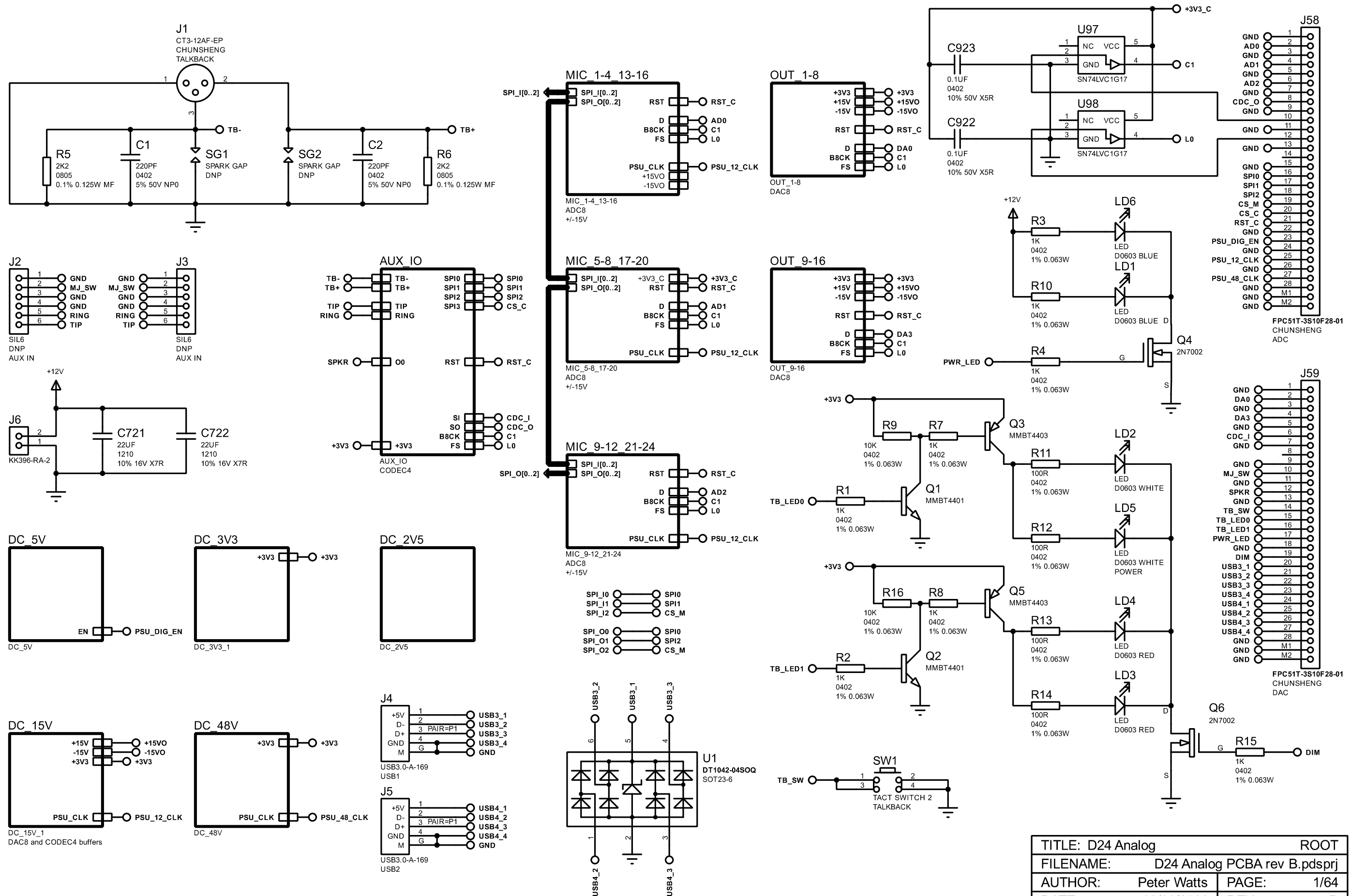
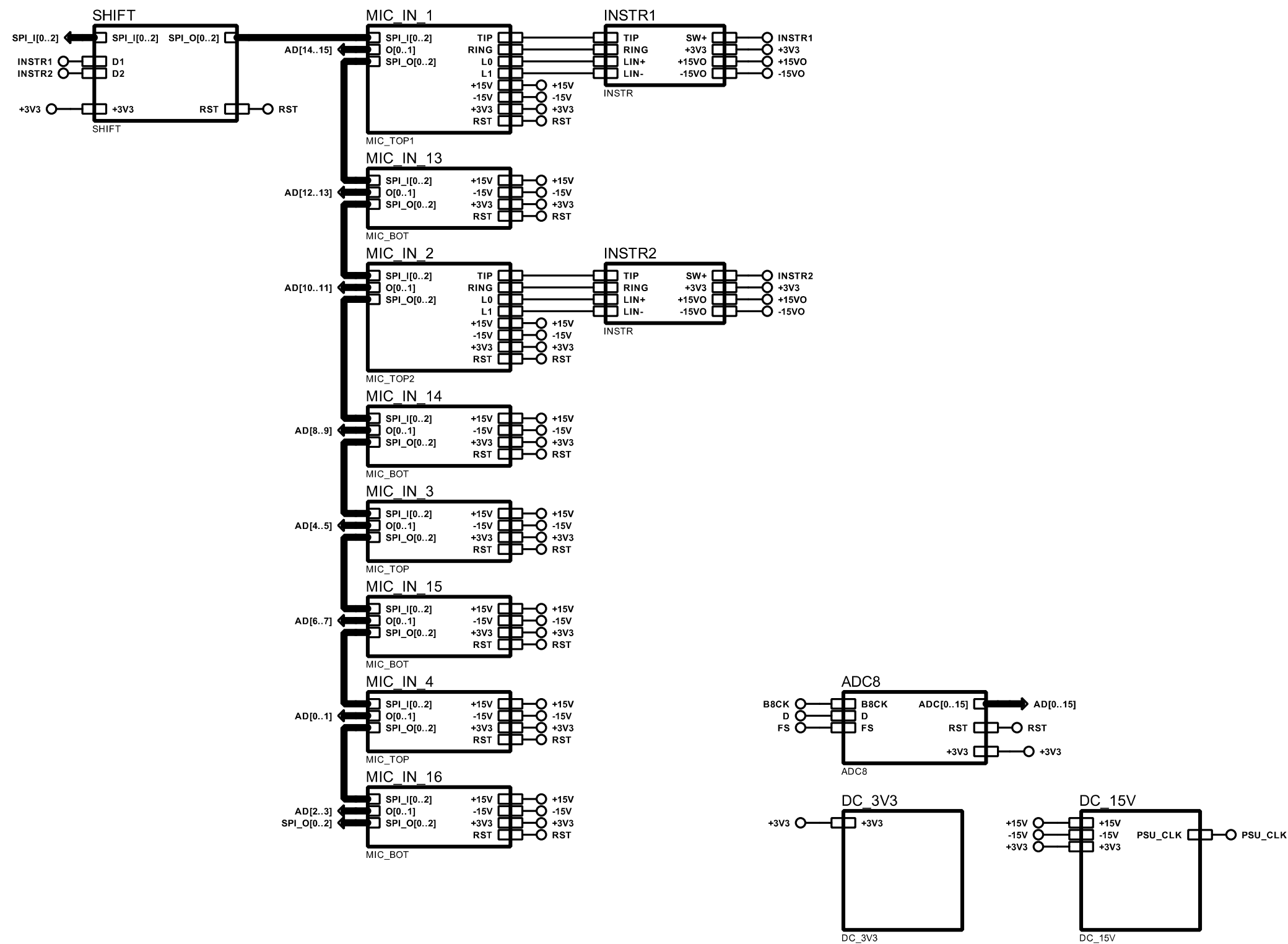
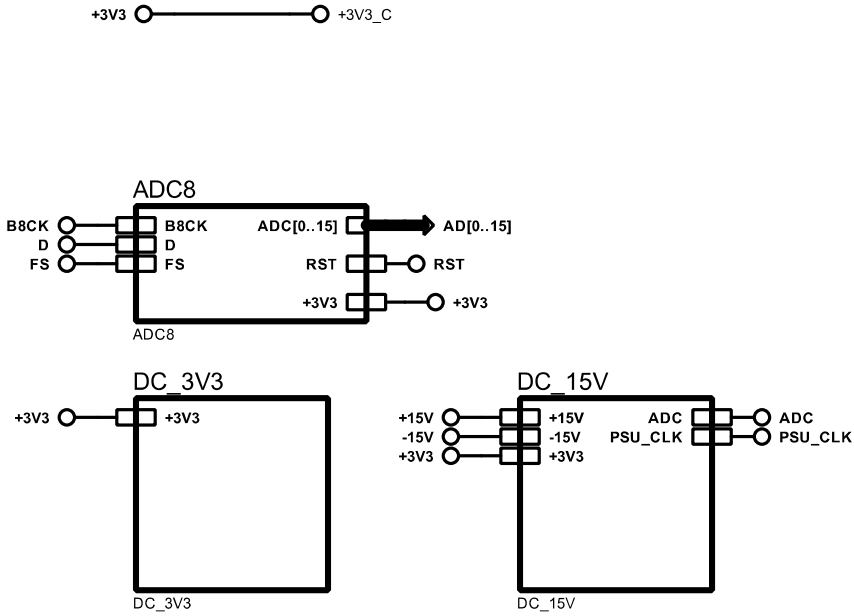
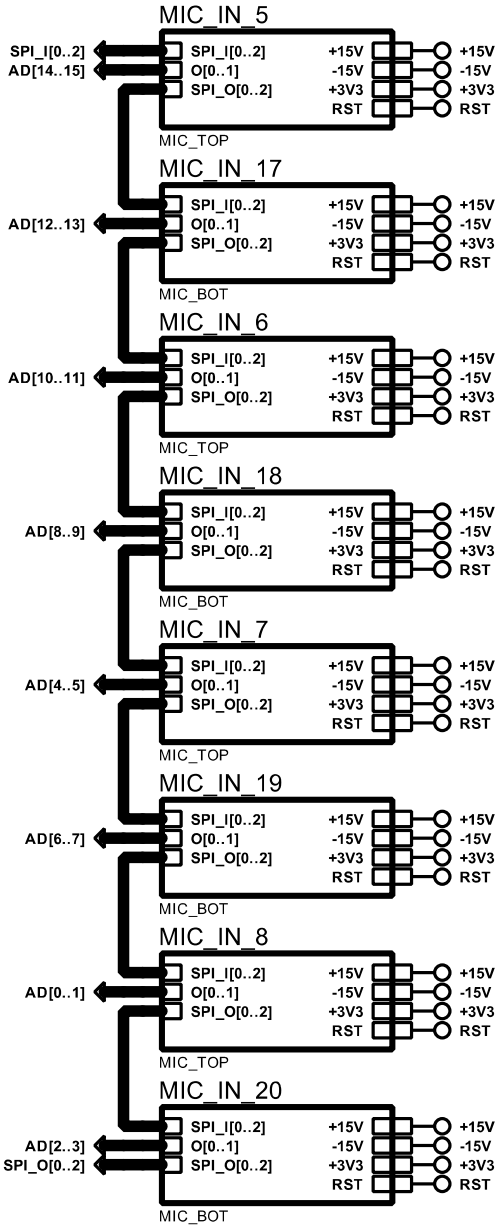




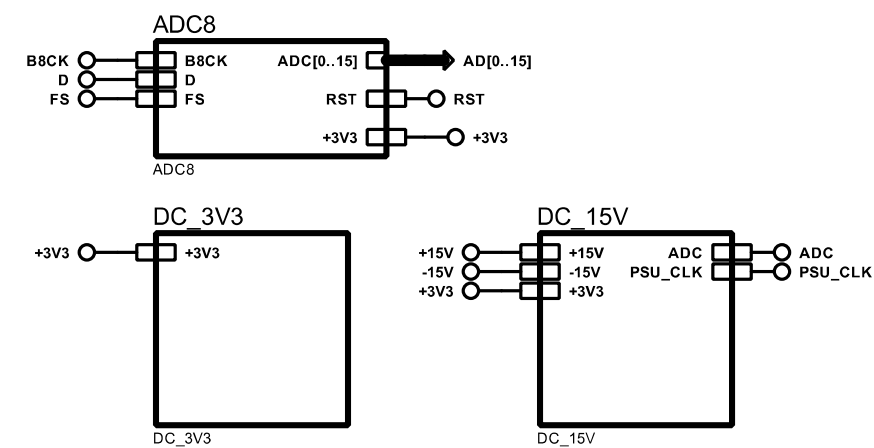
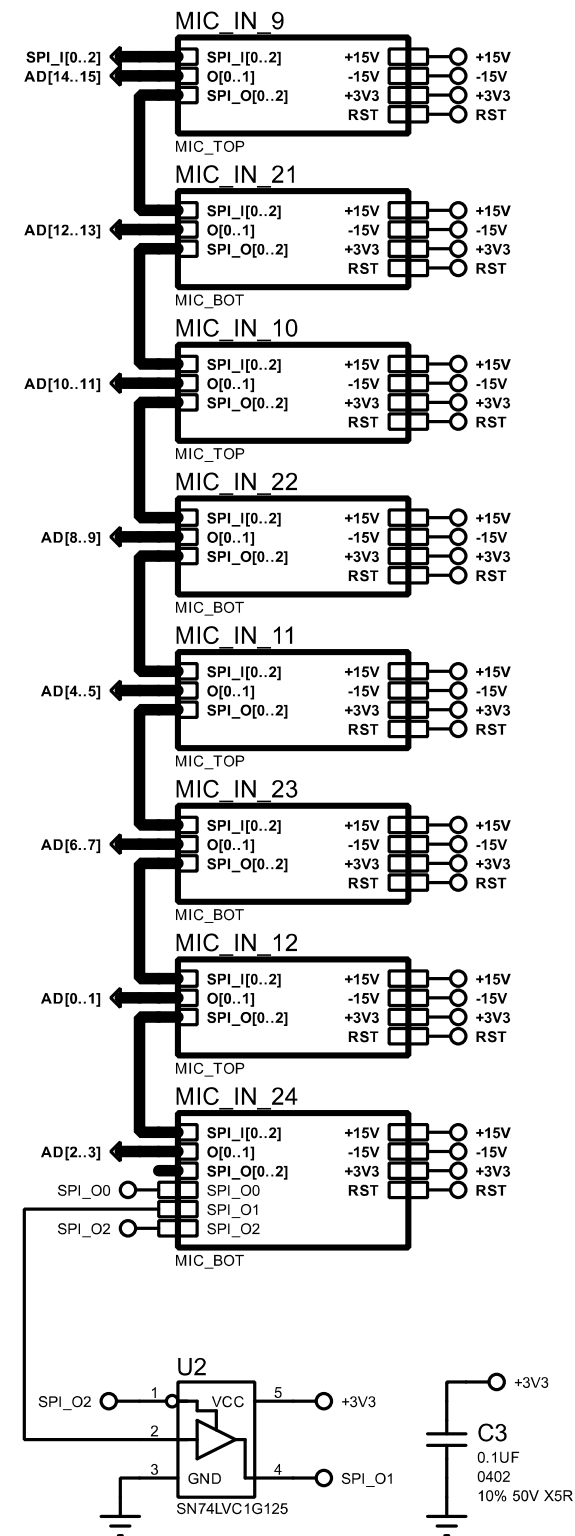
TITLE: D24 Analog		ROOT	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 1/64
DATE:		02/03/2026	REV: B



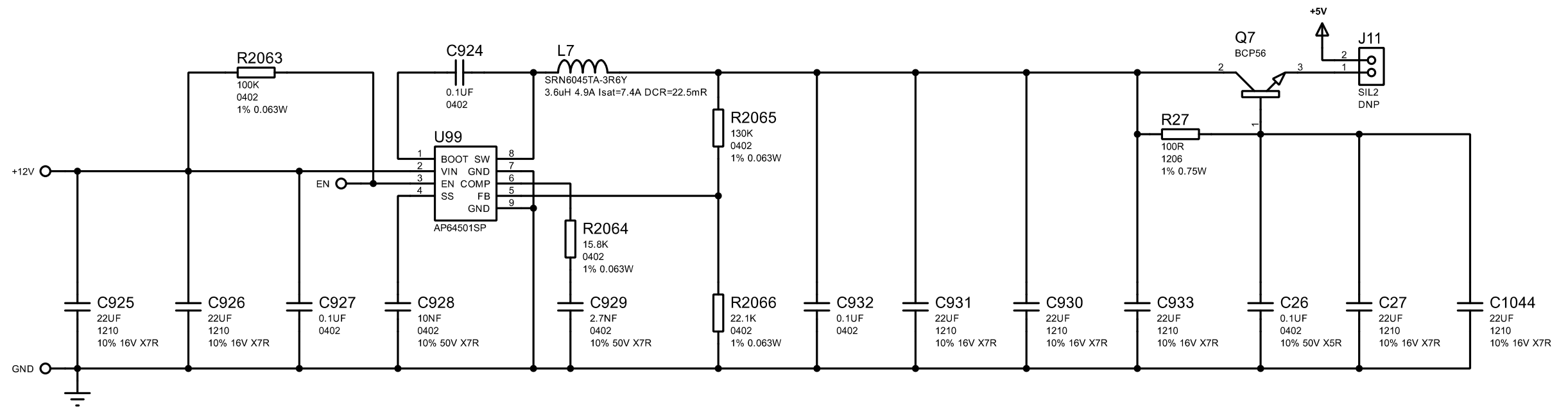
TITLE: D24 Analog		MIC_1-4_13-16	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 2/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		MIC_5-8_17-20	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 3/64
DATE:		02/03/2026	REV: B



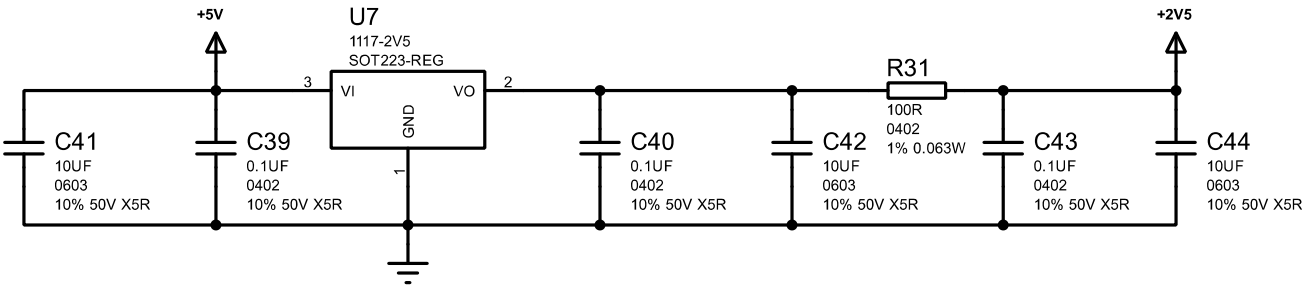
TITLE: D24 Analog		MIC_9-12_21-24	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 4/64
DATE:		02/03/2026	REV: B



RES VALUES FOR OUTPUT VOLTAGES
INPUT VOLTAGE 3.8V - 40V

OUTPUT	R16	R17
1.2V	3.74K	4.99K
1.5V	4.75K	8.66K
1.8V	5.62K	12.4K
2.5V	7.87K	21.5K
3.3V	10.5K	31.6K
5.0V	15.8K	52.3K
12V	37.4K	140K

TITLE: D24 Analog		DC_5V	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 5/64
DATE:		02/03/2026	REV: B

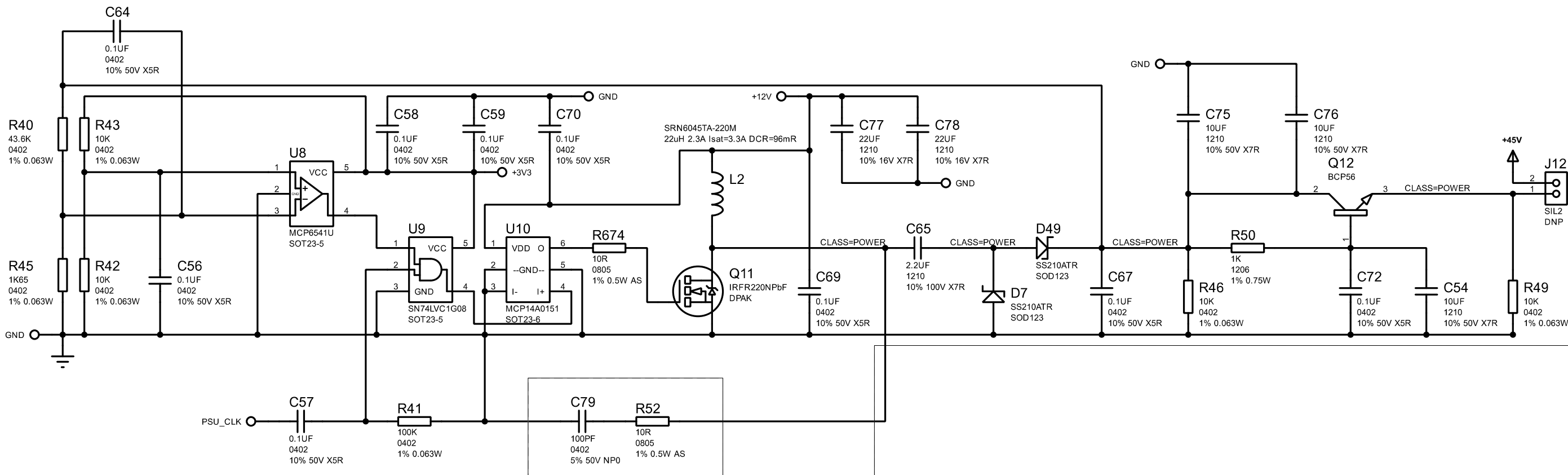


TITLE: D24 Analog		DC_2V5	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 6/64
DATE:		02/03/2026	REV: B

NOTES

gate res = 10R 0.5W metal

snubber 10R 1/8w as? 100pF C0G/NP0



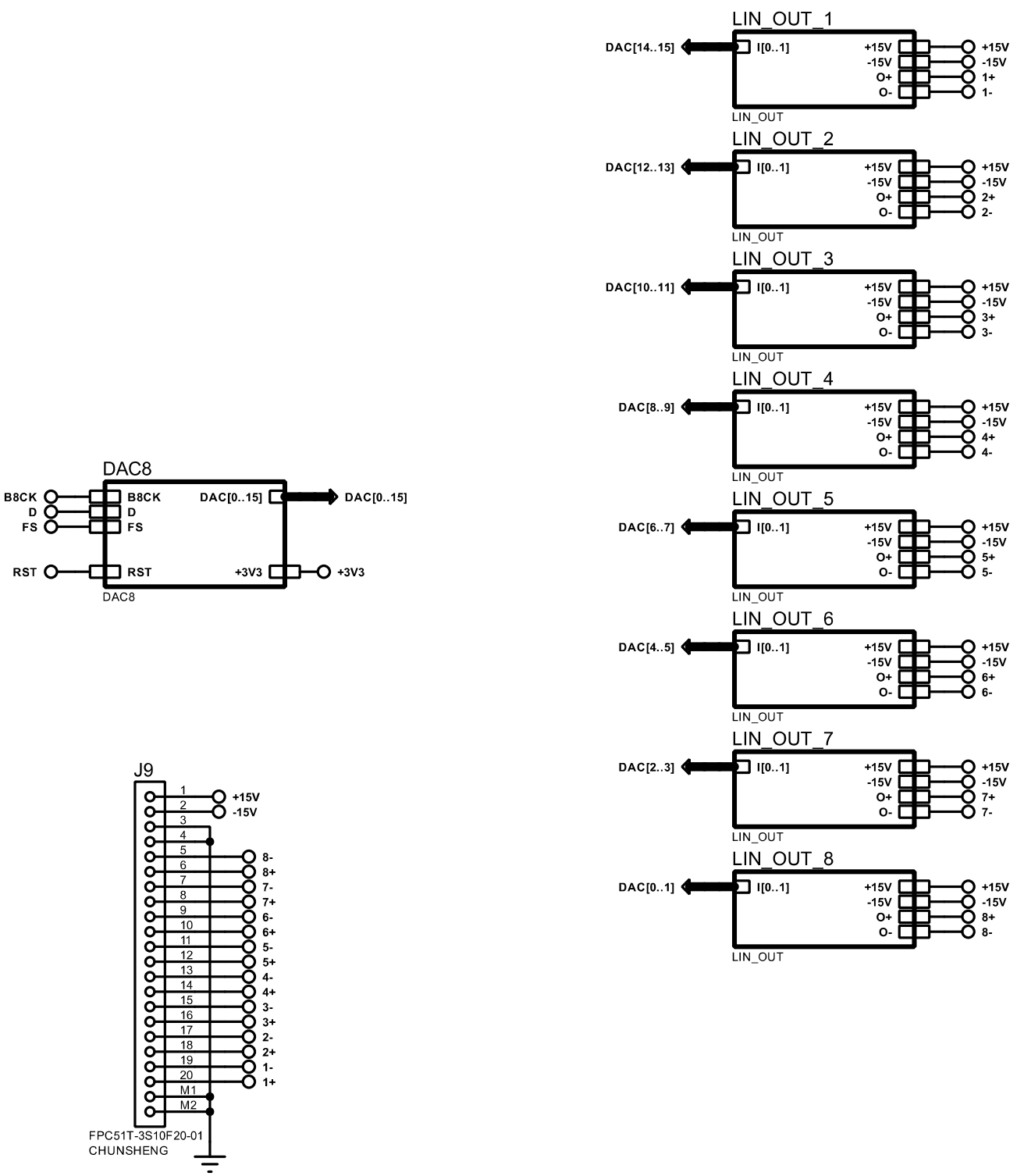
* optional snubber

NOTES

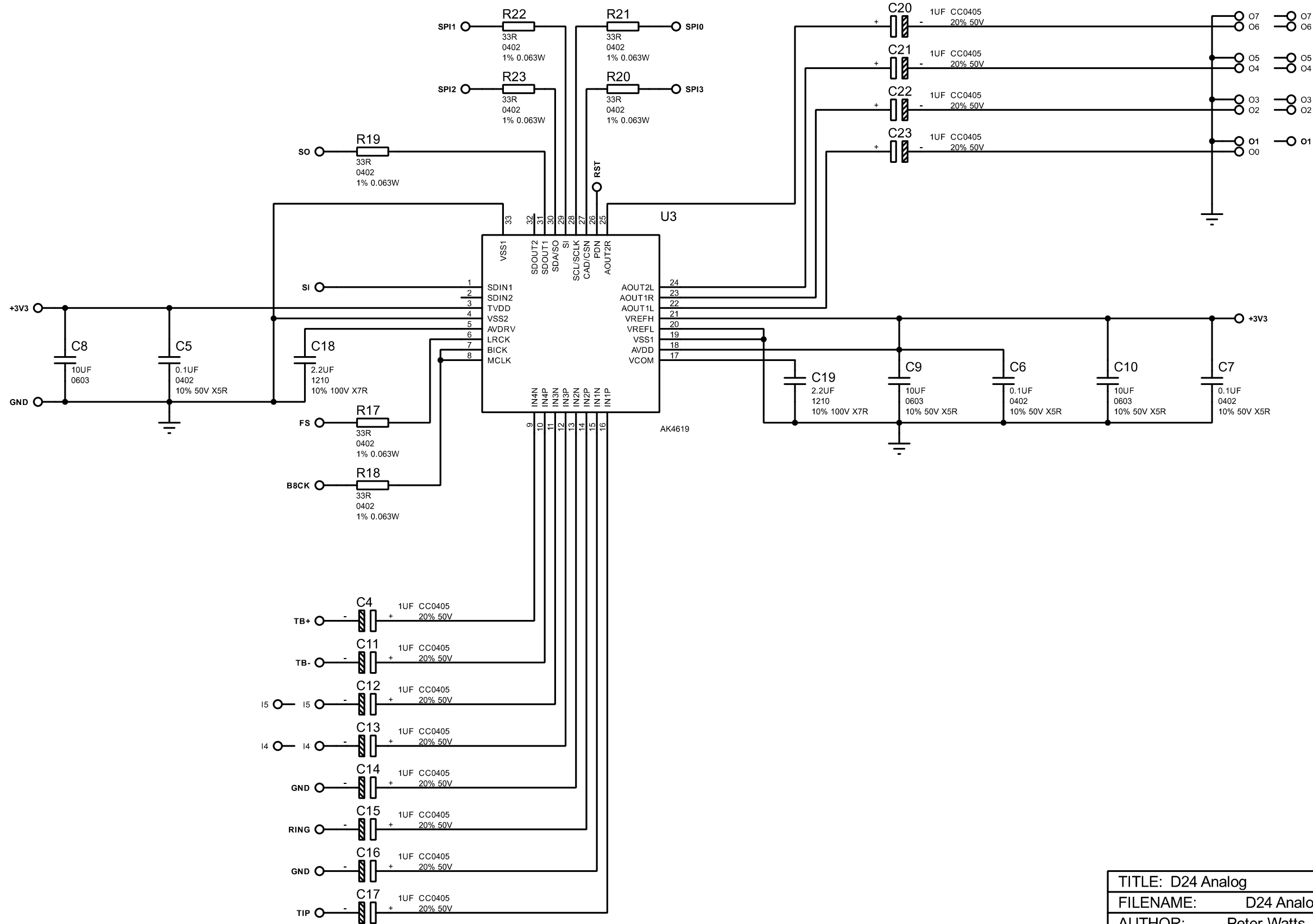
This module can also be used for 48V supply.
Change R2 to 46K35.
Change all 22uF caps to 10uF 50V.
Remove negative rail components.

* remove these components for 48V use

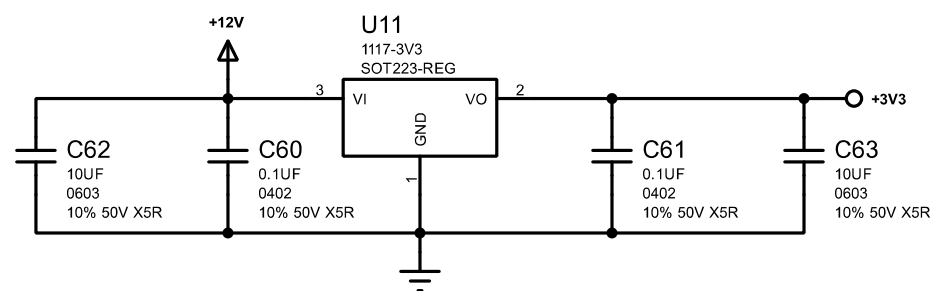
TITLE: D24 Analog		DC_48V	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 7/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		OUT_1-8	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 8/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		AUX_IO	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 9/64
DATE:		02/03/2026	REV: B

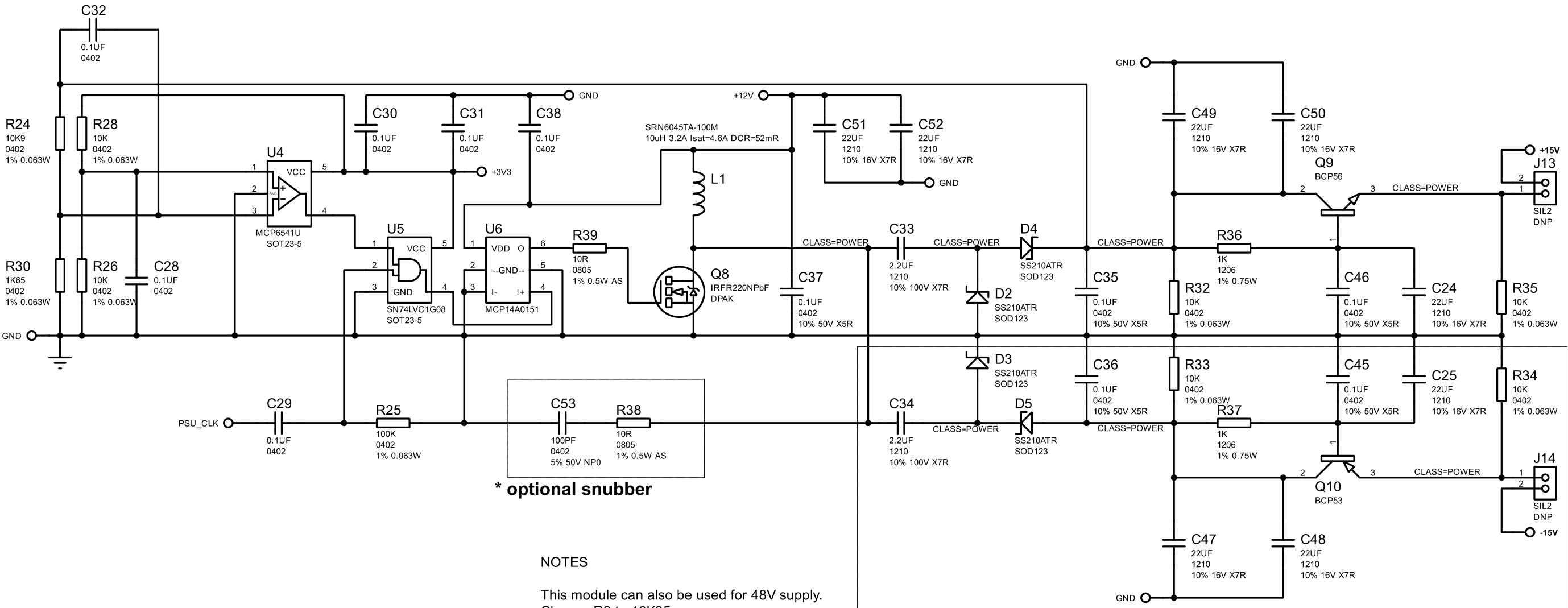


TITLE: D24 Analog		DC_3V3	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR: Peter Watts		PAGE: 10/64	
DATE: 02/03/2026		REV: B	

NOTES

gate res = 10R 0.5W metal

snubber 10R 1/8w as? 100pF C0G/NP0



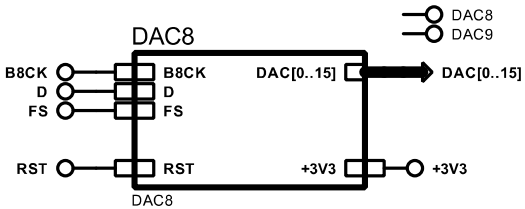
* optional snubber

NOTES

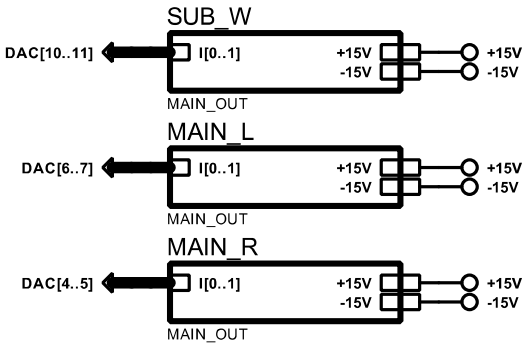
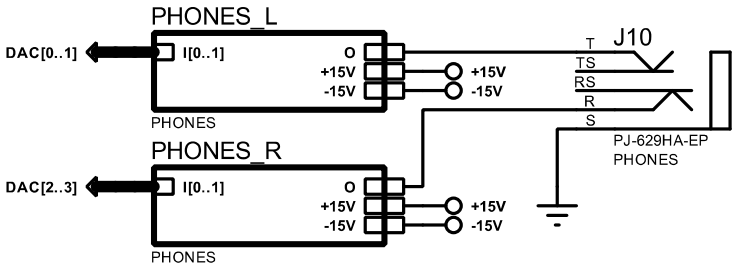
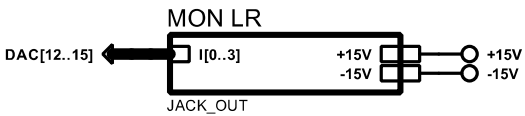
This module can also be used for 48V supply.
Change R2 to 46K35.
Change all 22UF caps to 10UF 50V.
Remove negative rail components.

* remove these components for 48V use

TITLE: D24 Analog		DC_15V	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 11/64
DATE:		02/03/2026	REV: B



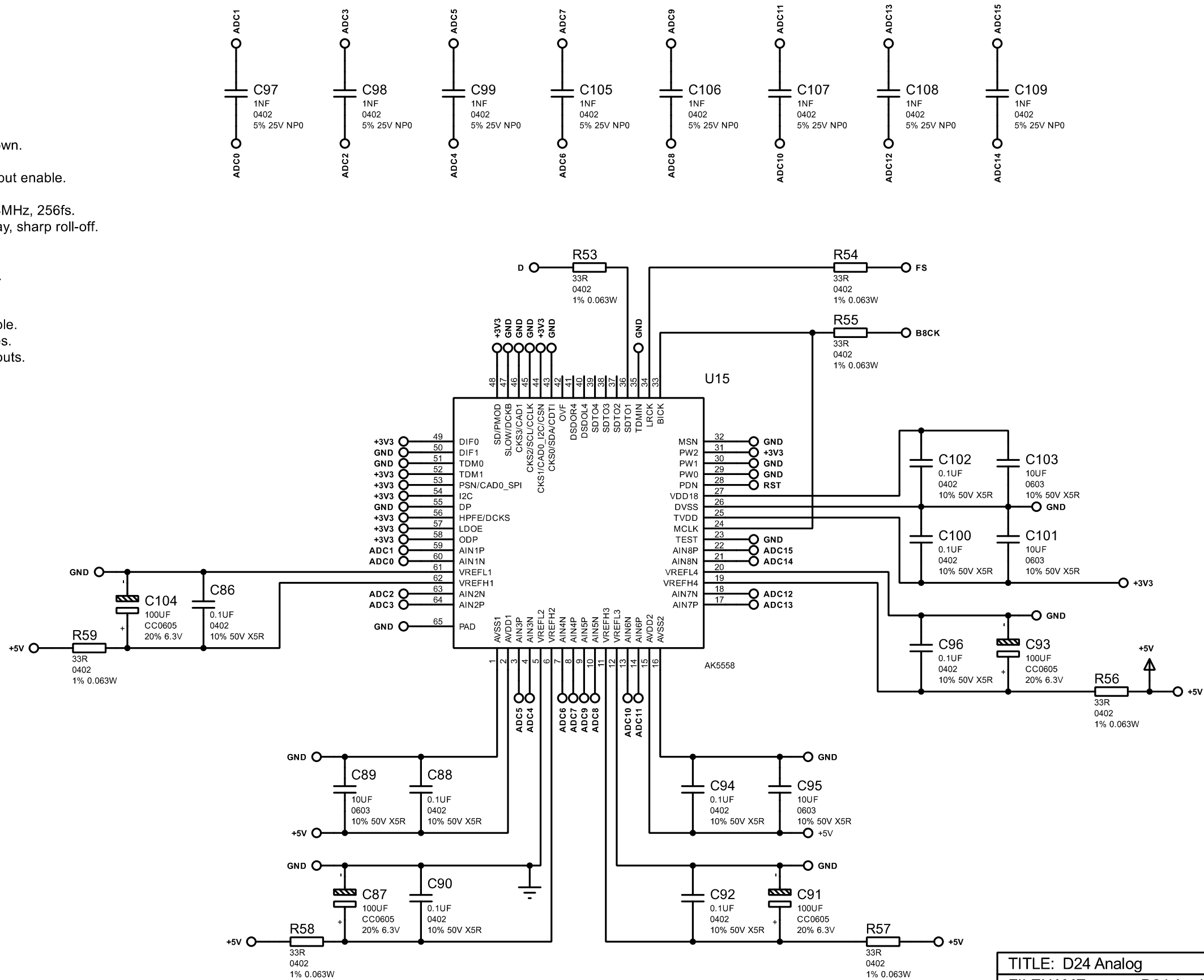
DAC8
DAC9



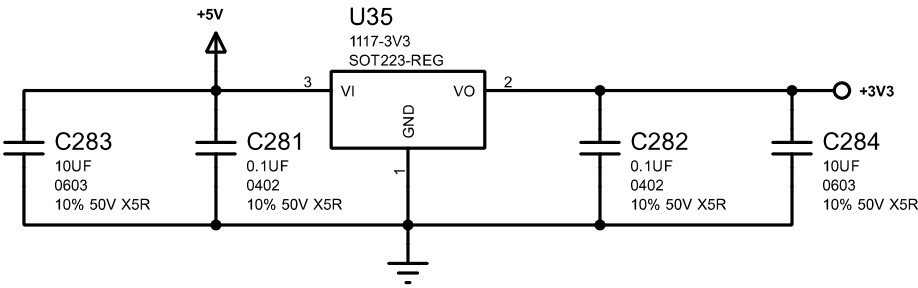
TITLE: D24 Analog		OUT_9-16	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 12/64
DATE:		02/03/2026	REV: B

Notes:
Place all 22R close to source.

PINS:
TEST = 0 = internal 100k pull-down.
MCLK = 12.288MHz.
PW0-2 = 001 = Channel 1-8 output enable.
MSN = 0 = Slave mode.
CKS0-3 = 0100 = 48kHz, 12.288MHz, 256fs.
SLOW/SD = 01 = HPF short delay, sharp roll-off.
DIF0-1 = 10 = I2S 24-bit mode.
TDM0-1 = 01 = I2S 24-bit mode.
PSN = 1 = Parallel control mode.
I2C = 1 = Parallel control mode.
DP = 0 = PCM mode.
HPFE = 1 = High pass filter enable.
LDOE = 1 = Enable LDO voltages.
ODP = 1 = Normal TDM128 outputs.



TITLE: D24 Analog		ADC8	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 13/64
DATE:		02/03/2026	REV: B

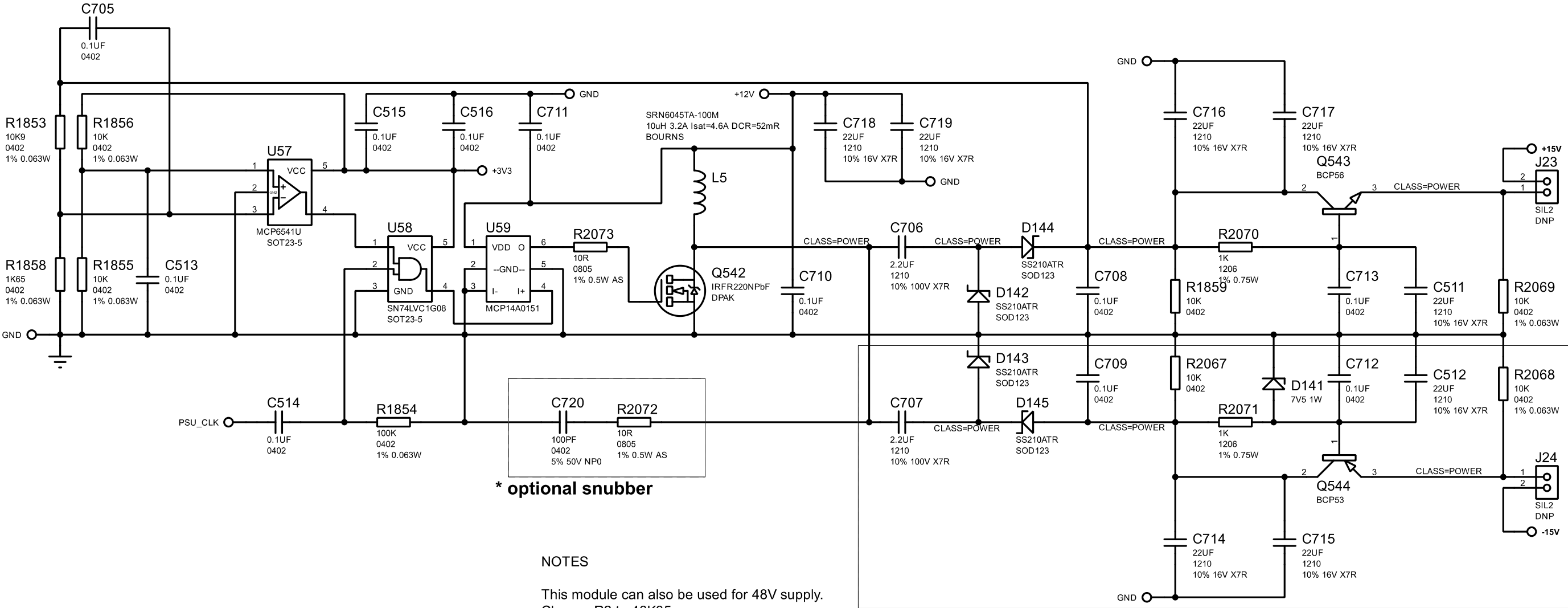


TITLE: D24 Analog		DC_3V3	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 14/64
DATE:		02/03/2026	REV: B

NOTES

gate res = 10R 0.5W metal

snubber 10R 1/8w as? 100pF C0G/NP0



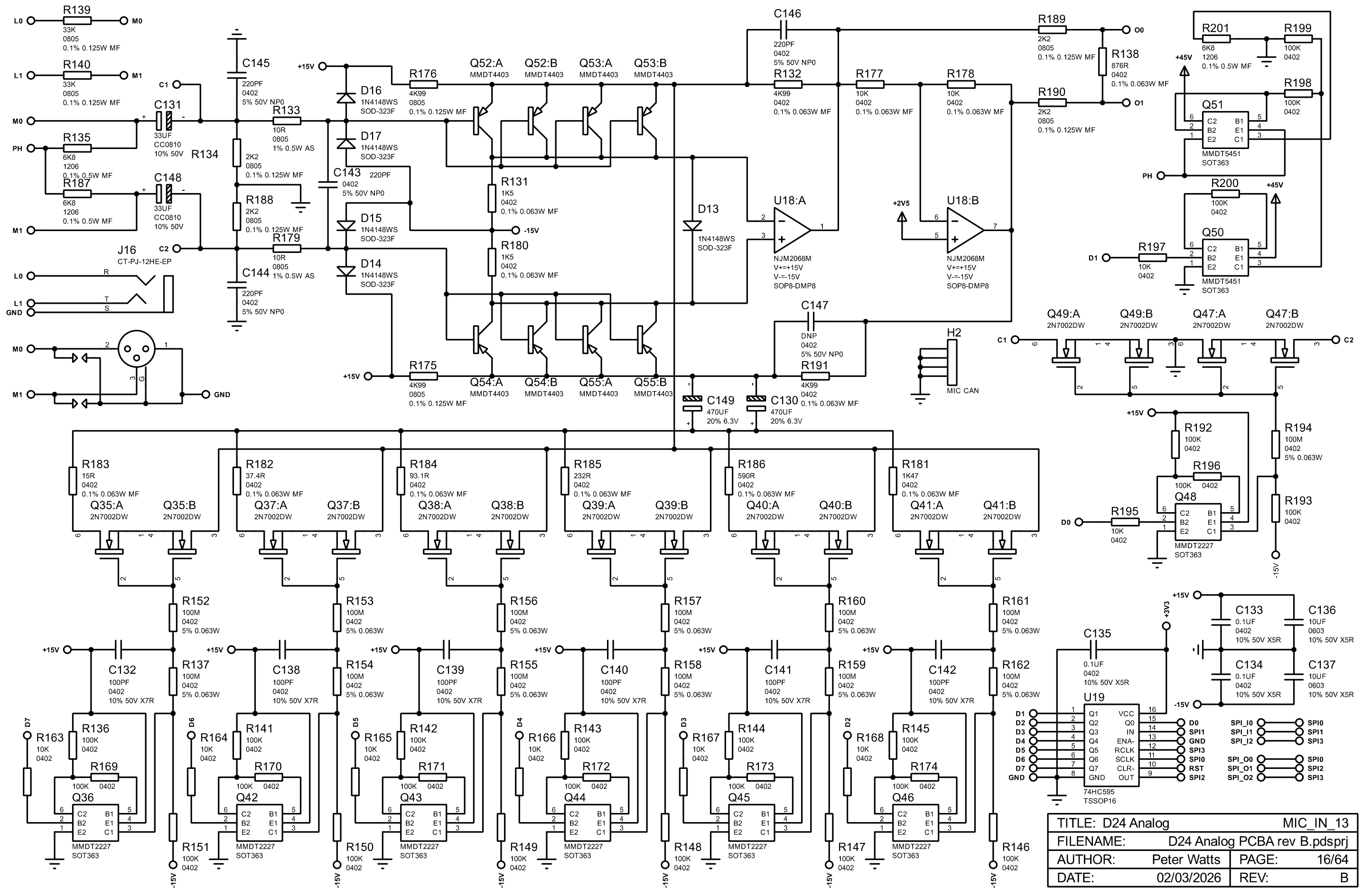
* optional snubber

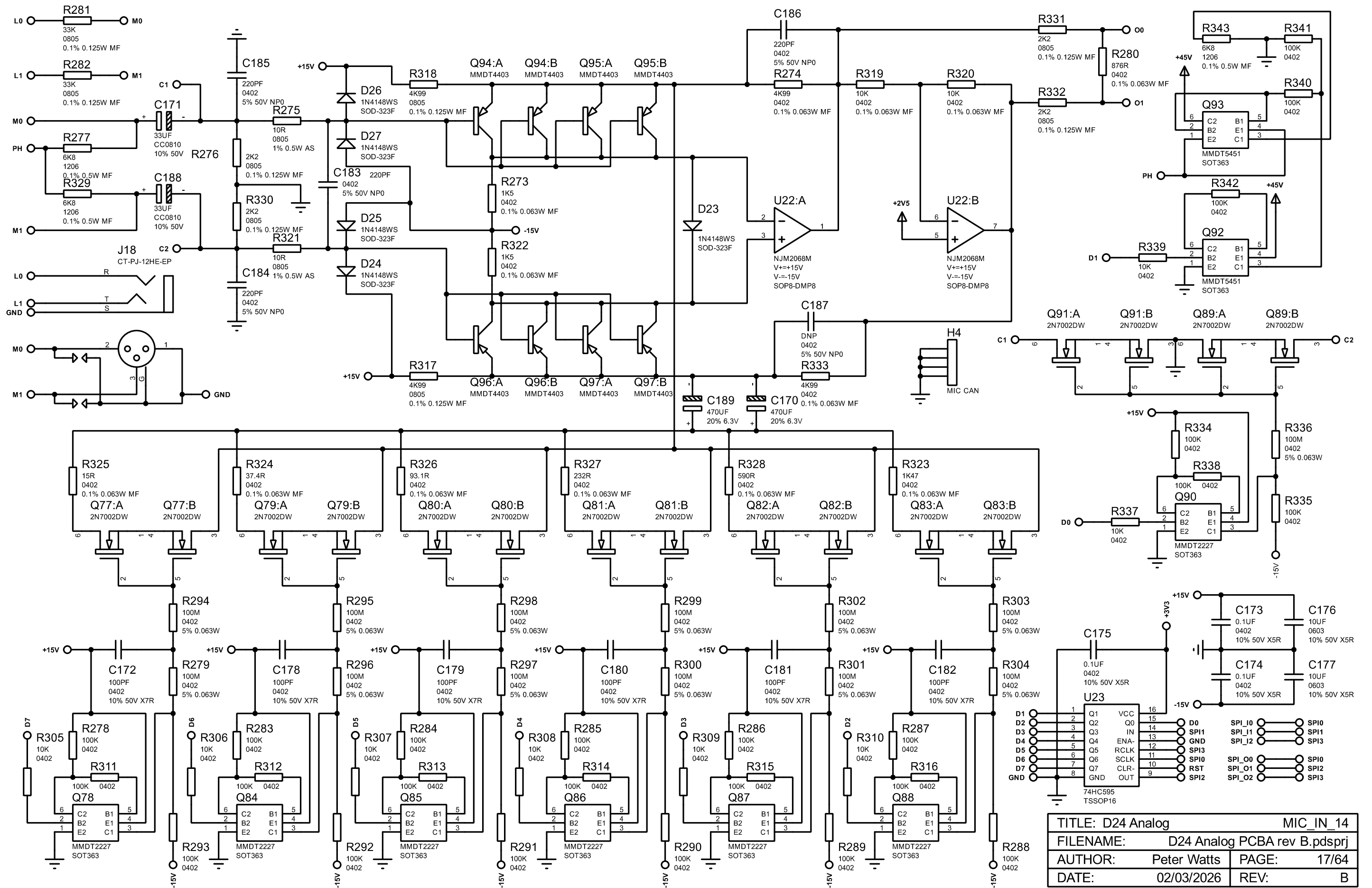
NOTES

This module can also be used for 48V supply.
Change R2 to 46K35.
Change all 22UF caps to 10UF 50V.
Remove negative rail components.

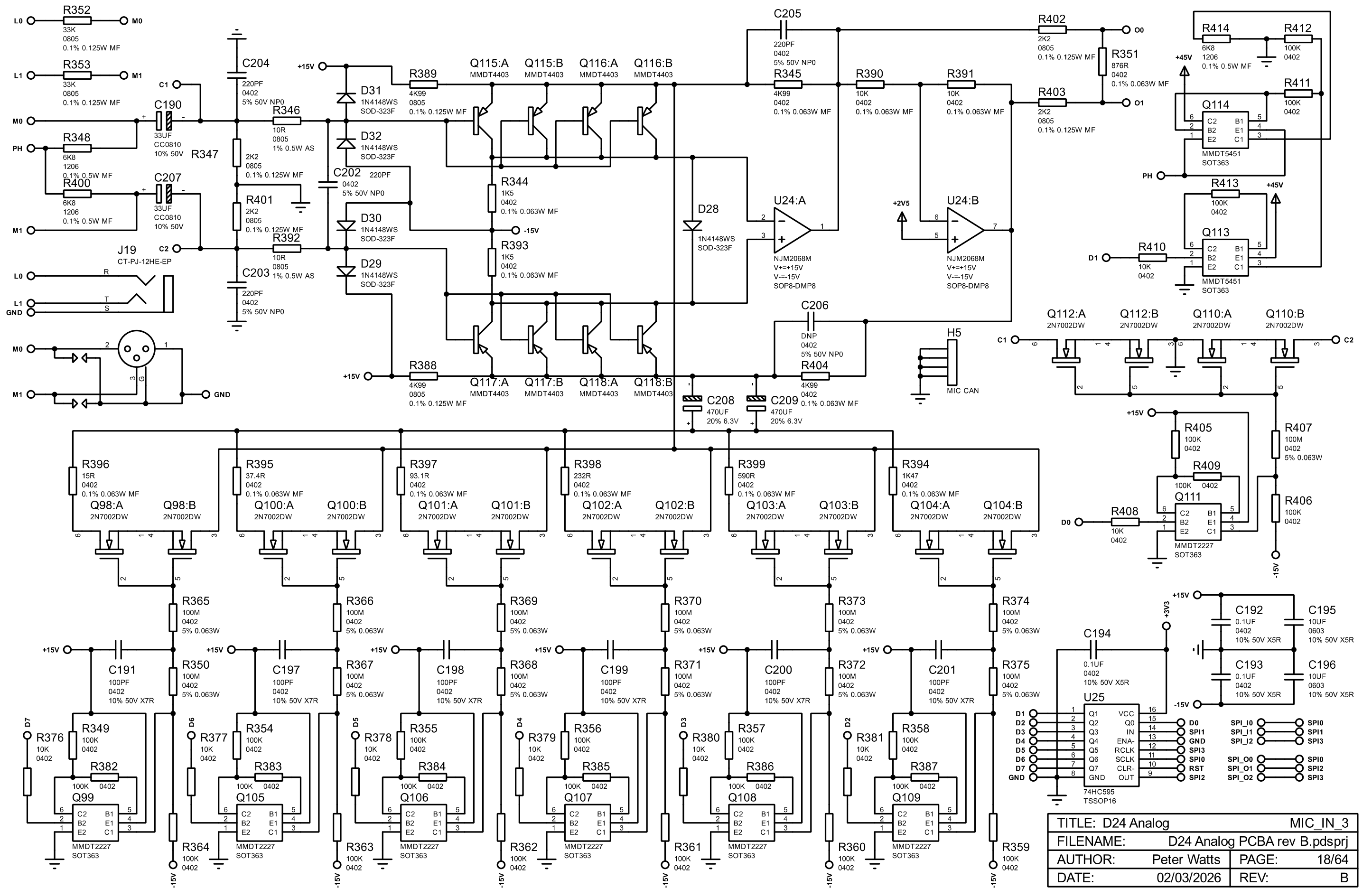
* remove these components for 48V use

TITLE: D24 Analog		DC_15V	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 15/64
DATE:		02/03/2026	REV: B

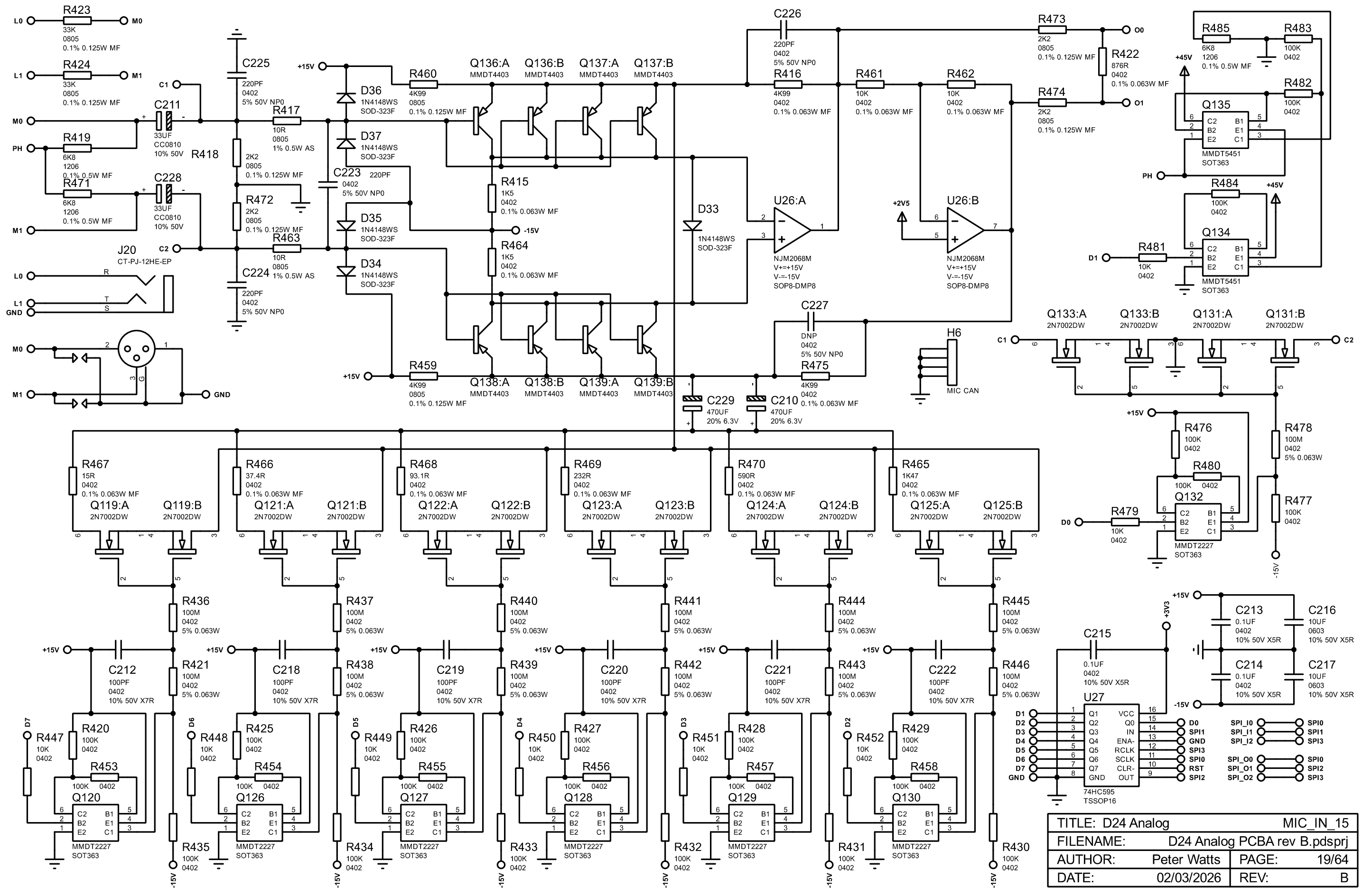




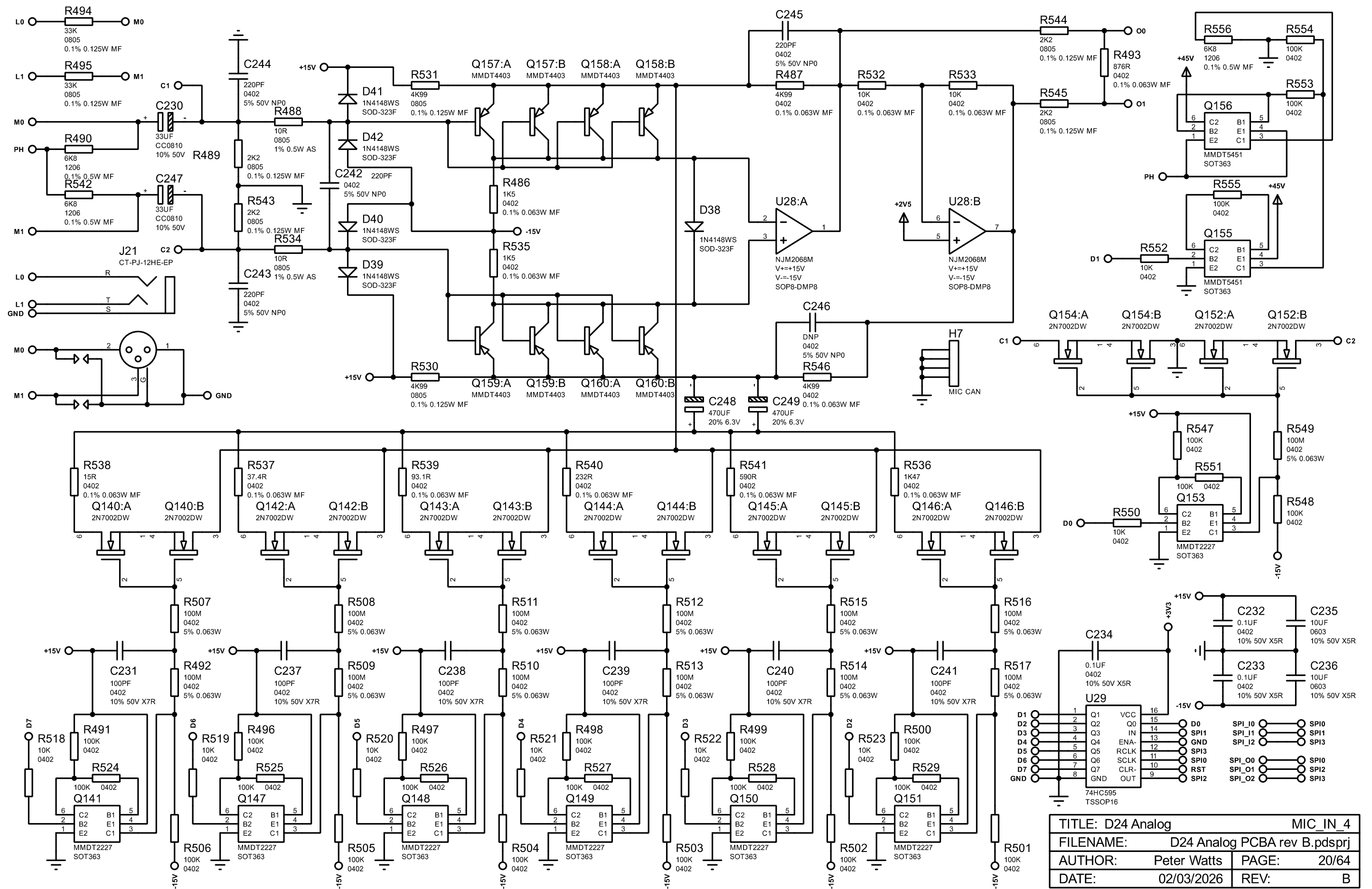
TITLE: D24 Analog		MIC_IN_14	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 17/64	
DATE: 02/03/2026		REV: B	

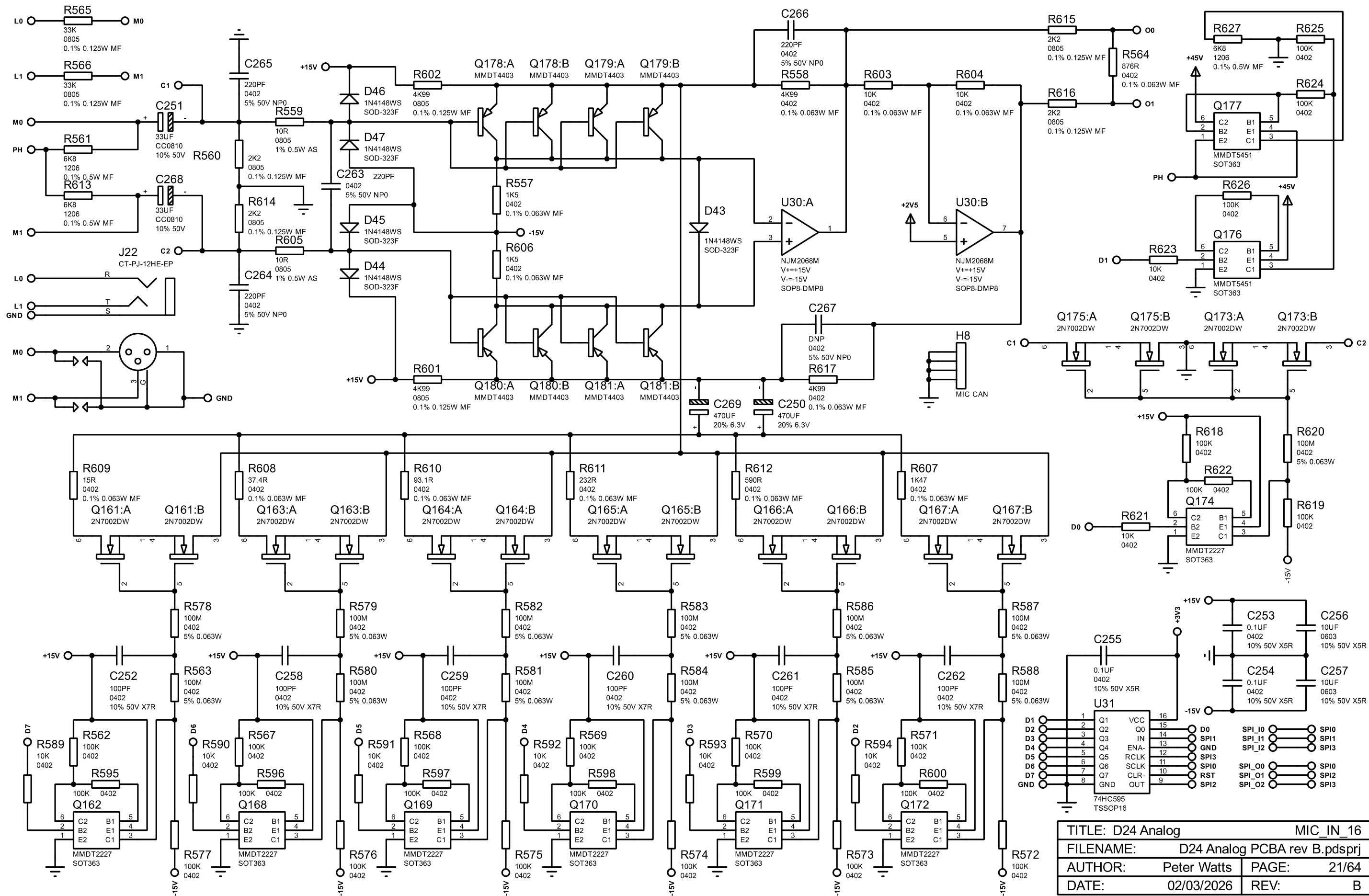


TITLE: D24 Analog		MIC_IN_3	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 18/64	
DATE: 02/03/2026		REV: B	

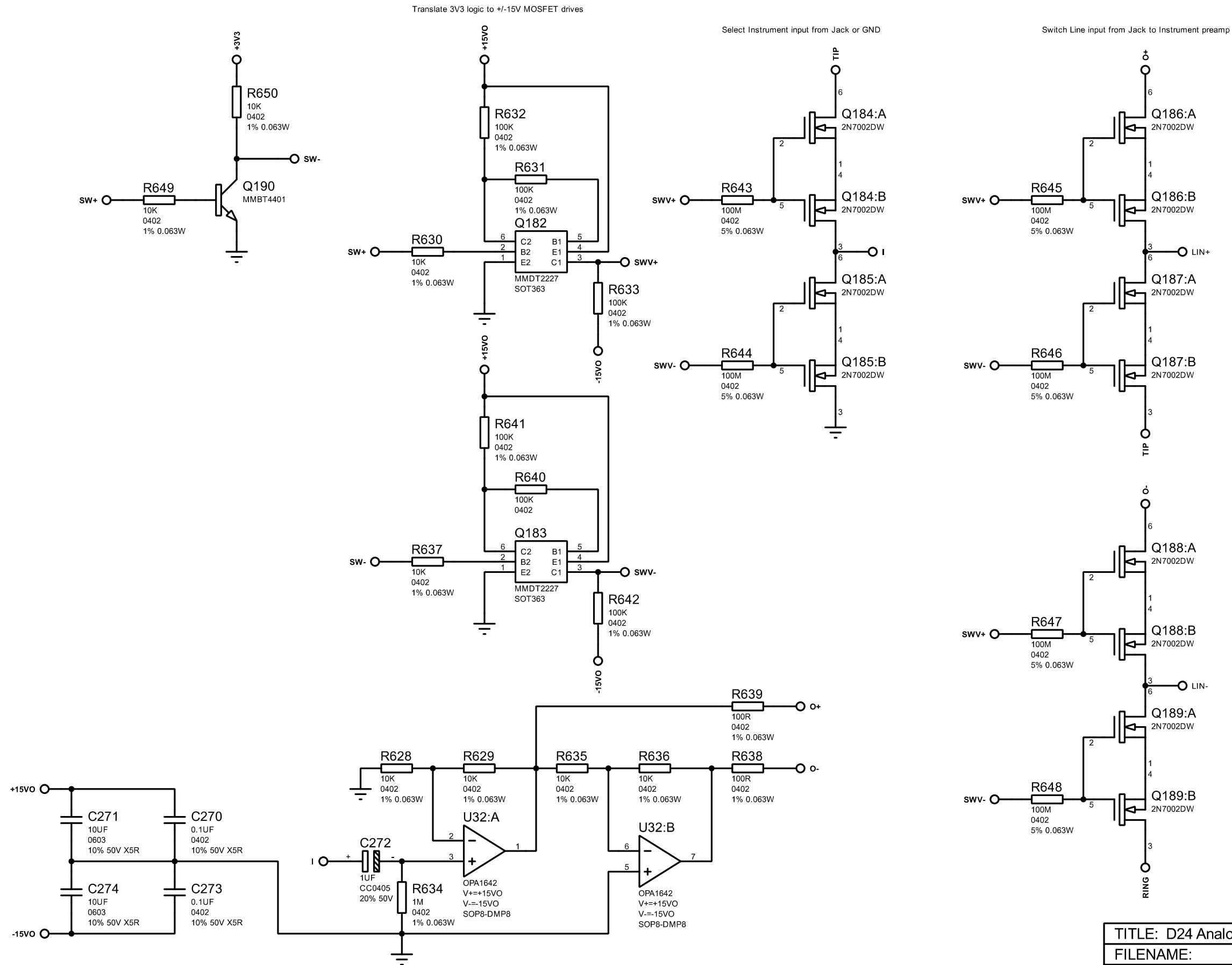


TITLE: D24 Analog		MIC_IN_15	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 19/64	
DATE: 02/03/2026		REV: B	

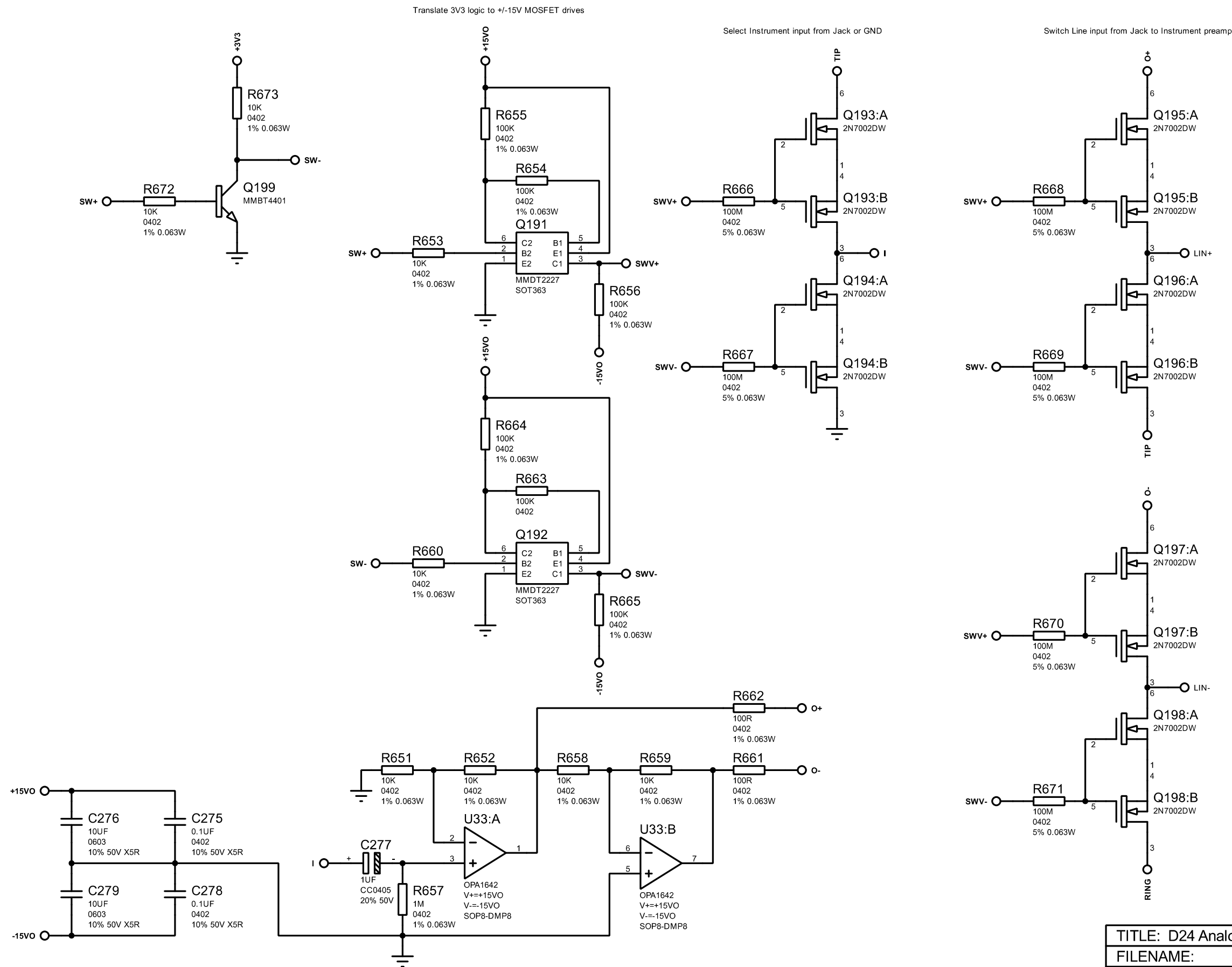




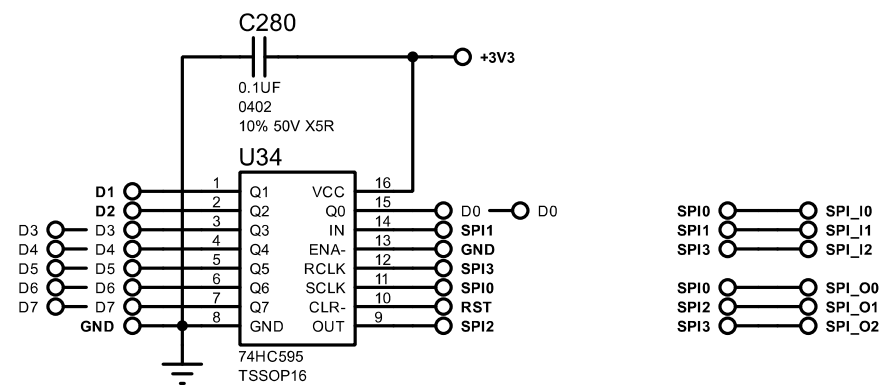
TITLE: D24 Analog		MIC_IN_16	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 21/64	
DATE: 02/03/2026		REV: B	



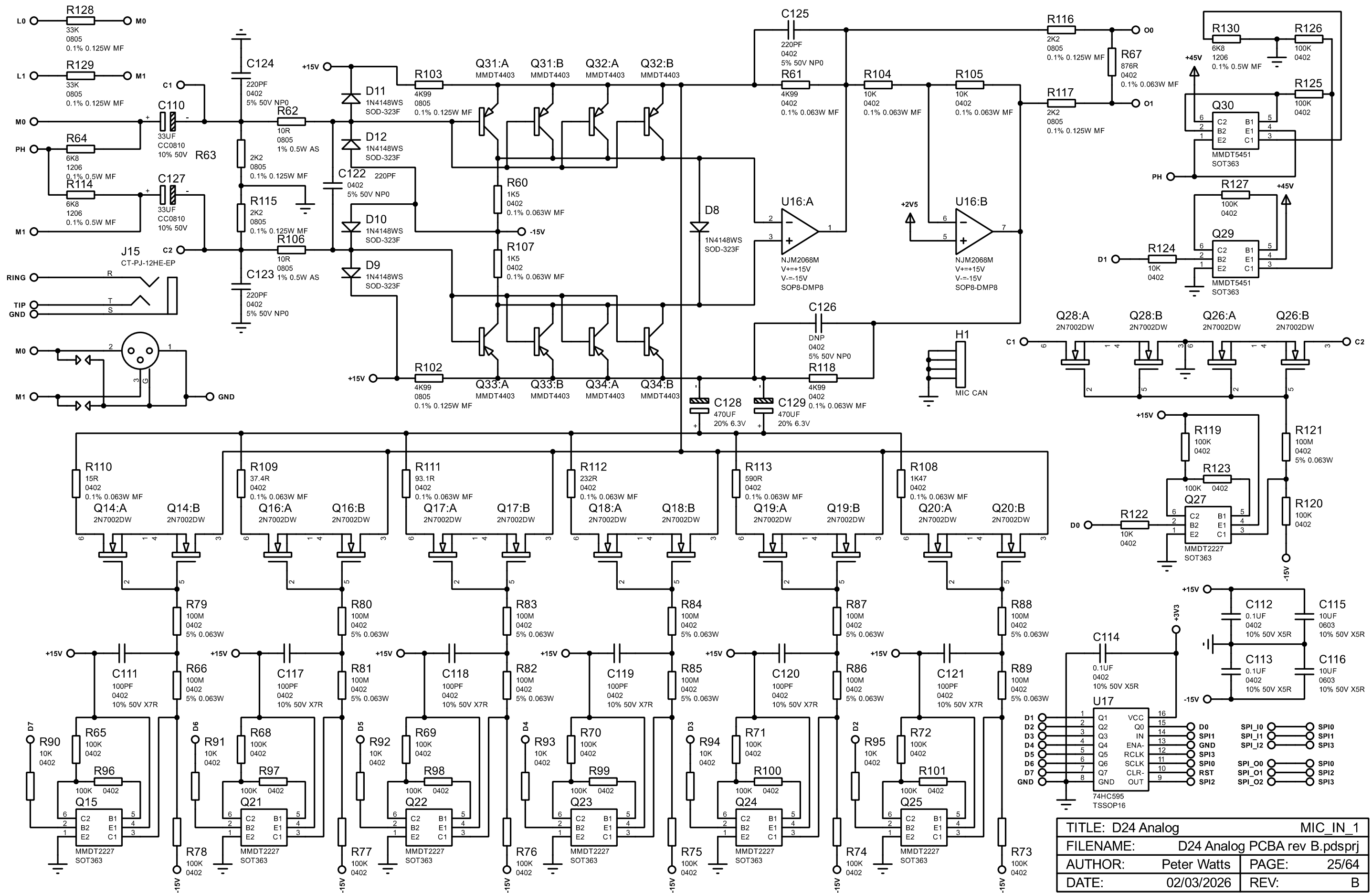
TITLE: D24 Analog		INSTR1	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 22/64
DATE:		02/03/2026	REV: B



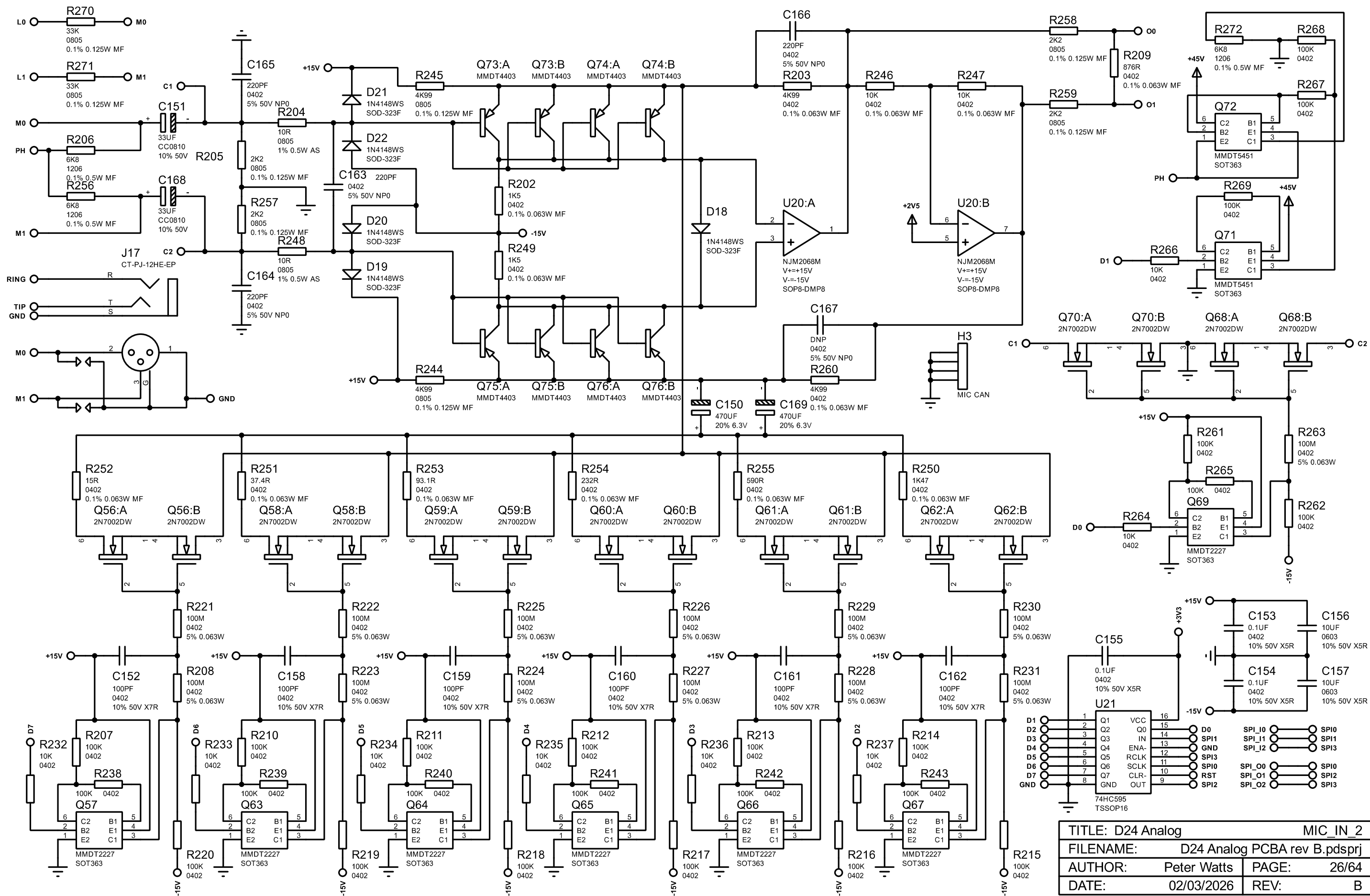
TITLE: D24 Analog		INSTR2	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 23/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		SHIFT	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 24/64
DATE:		02/03/2026	REV: B

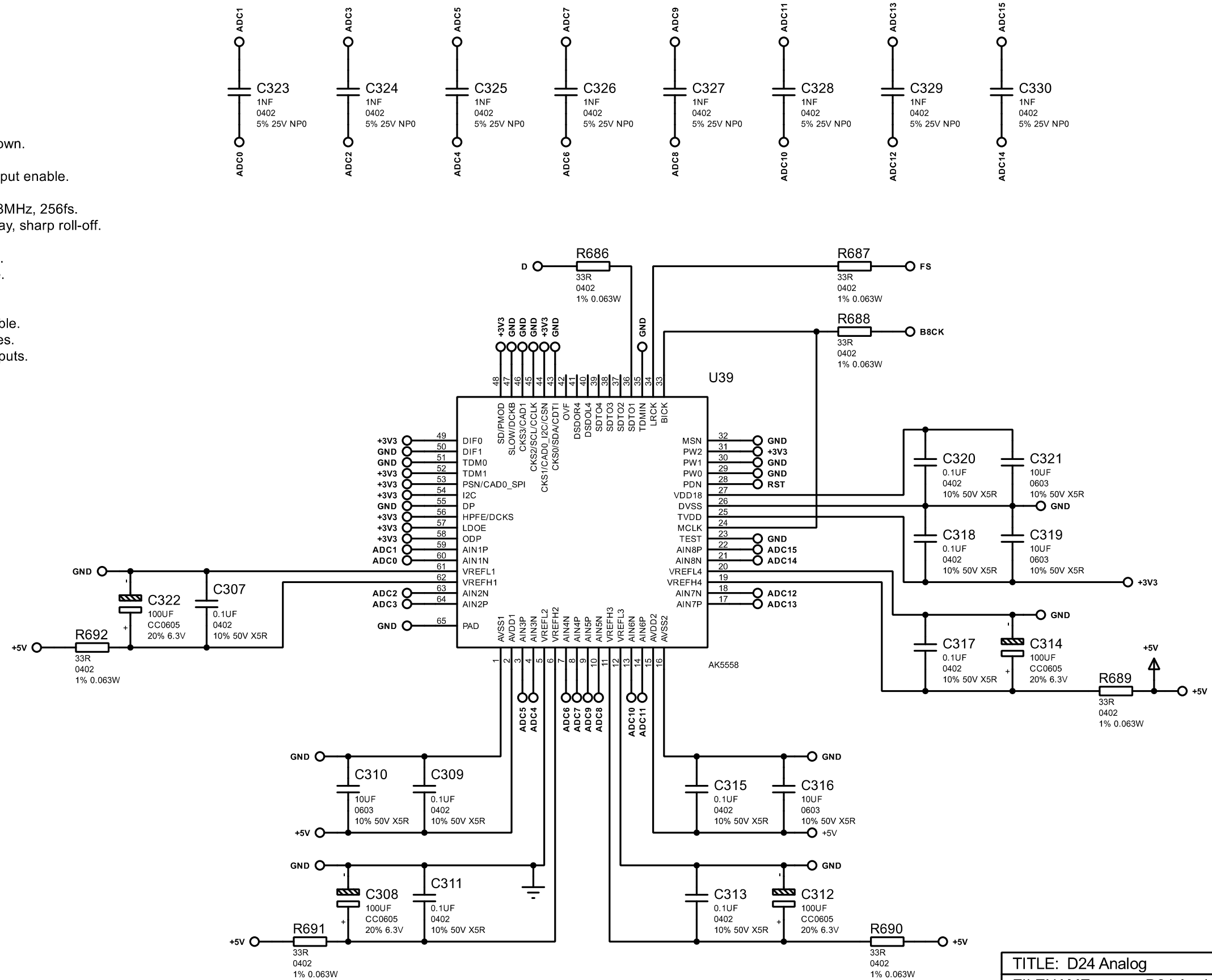


TITLE: D24 Analog		MIC_IN_1	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 25/64
DATE:		02/03/2026	REV: B

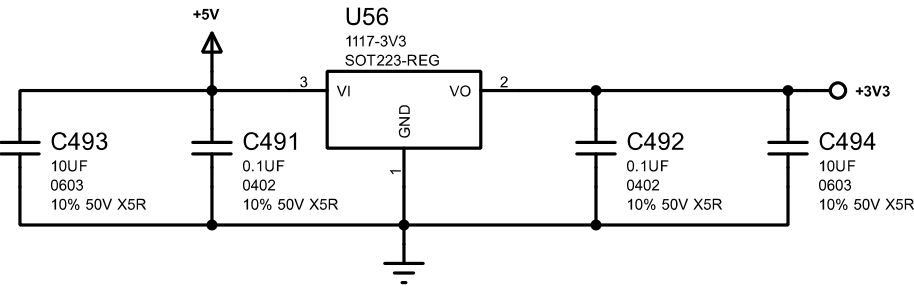


TITLE: D24 Analog		MIC_IN_2	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 26/64	
DATE: 02/03/2026		REV: B	

PINS:
TEST = 0 = internal 100k pull-down.
MCLK = 12.288MHz.
PW0-2 = 001 = Channel 1-8 output enable.
MSN = 0 = Slave mode.
CKS0-3 = 0100 = 48kHz, 12.288MHz, 256fs.
SLOW/SD = 01 = HPF short delay, sharp roll-off.
DIF0-1 = 10 = I2S 24-bit mode.
TDM0-1 = 01 = I2S 24-bit mode.
PSN = 1 = Parallel control mode.
I2C = 1 = Parallel control mode.
DP = 0 = PCM mode.
HPFE = 1 = High pass filter enable.
LDOE = 1 = Enable LDO voltages.
ODP = 1 = Normal TDM128 outputs.

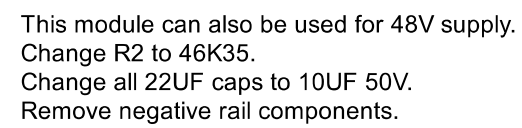


TITLE: D24 Analog		ADC8	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:	Peter Watts	PAGE:	27/64
DATE:	02/03/2026	REV:	B

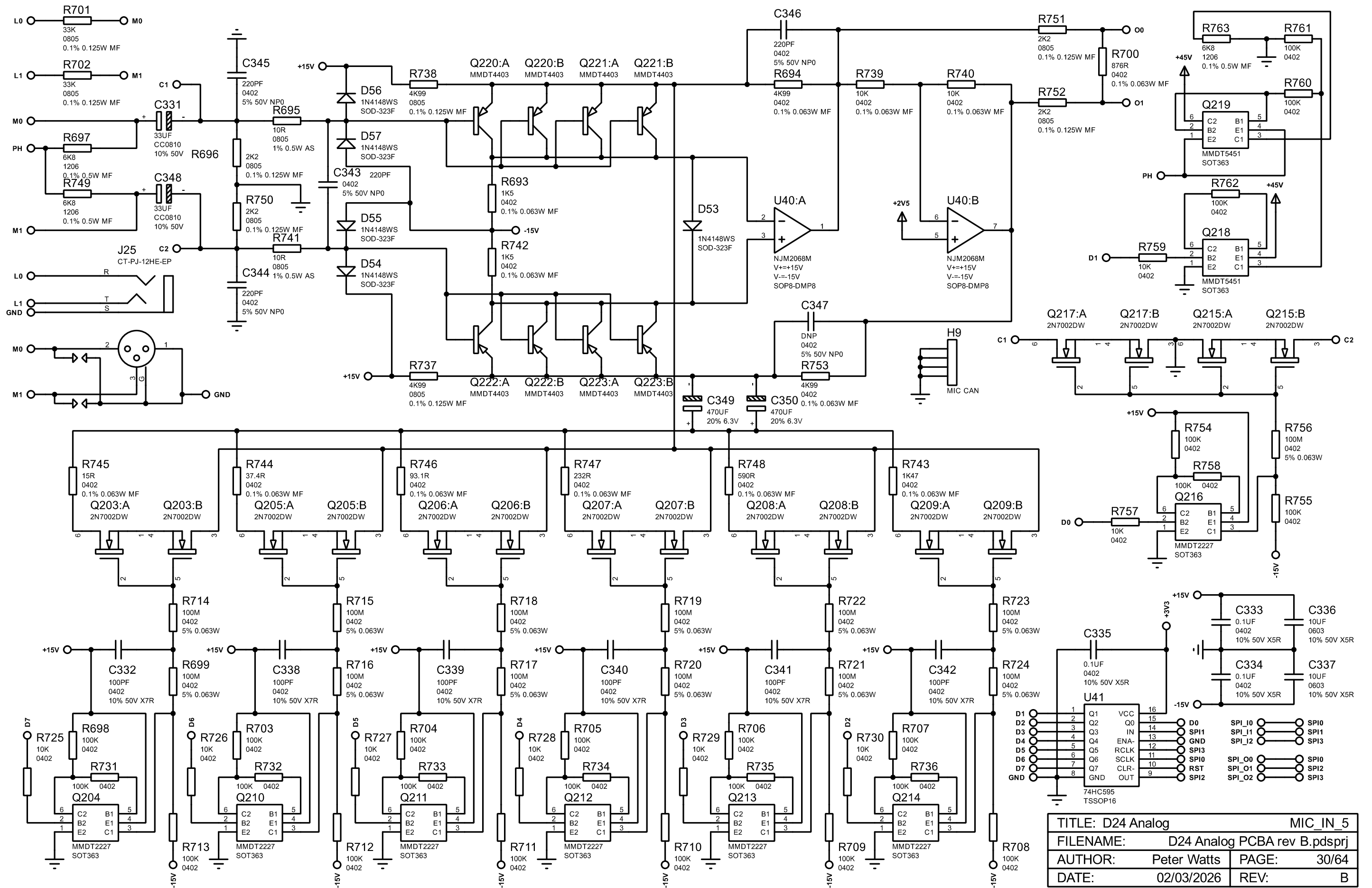


TITLE: D24 Analog		DC_3V3	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 28/64
DATE:		02/03/2026	REV: B

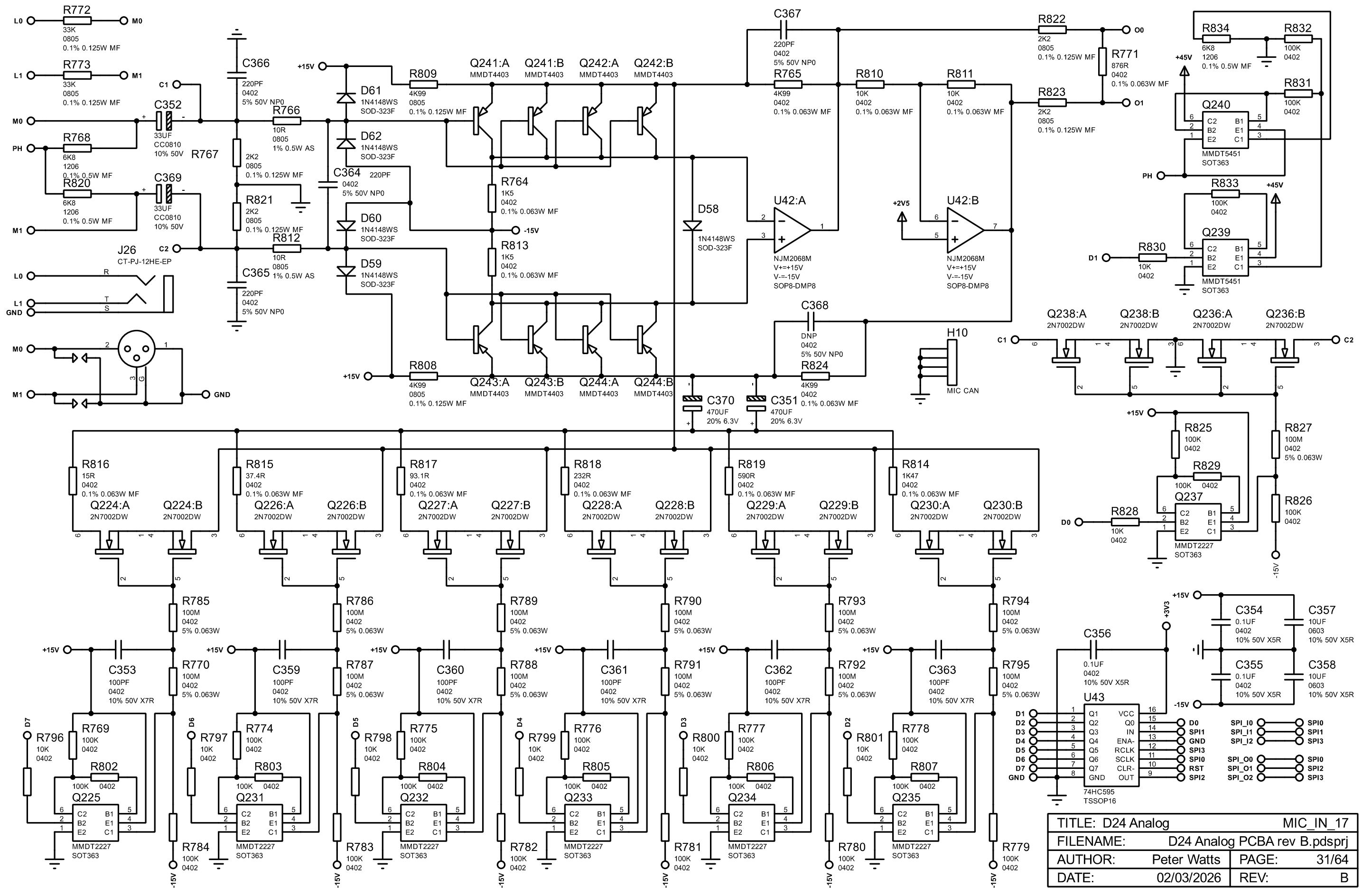
snubber 10R 1/8w as? 100pF C0G/NP0



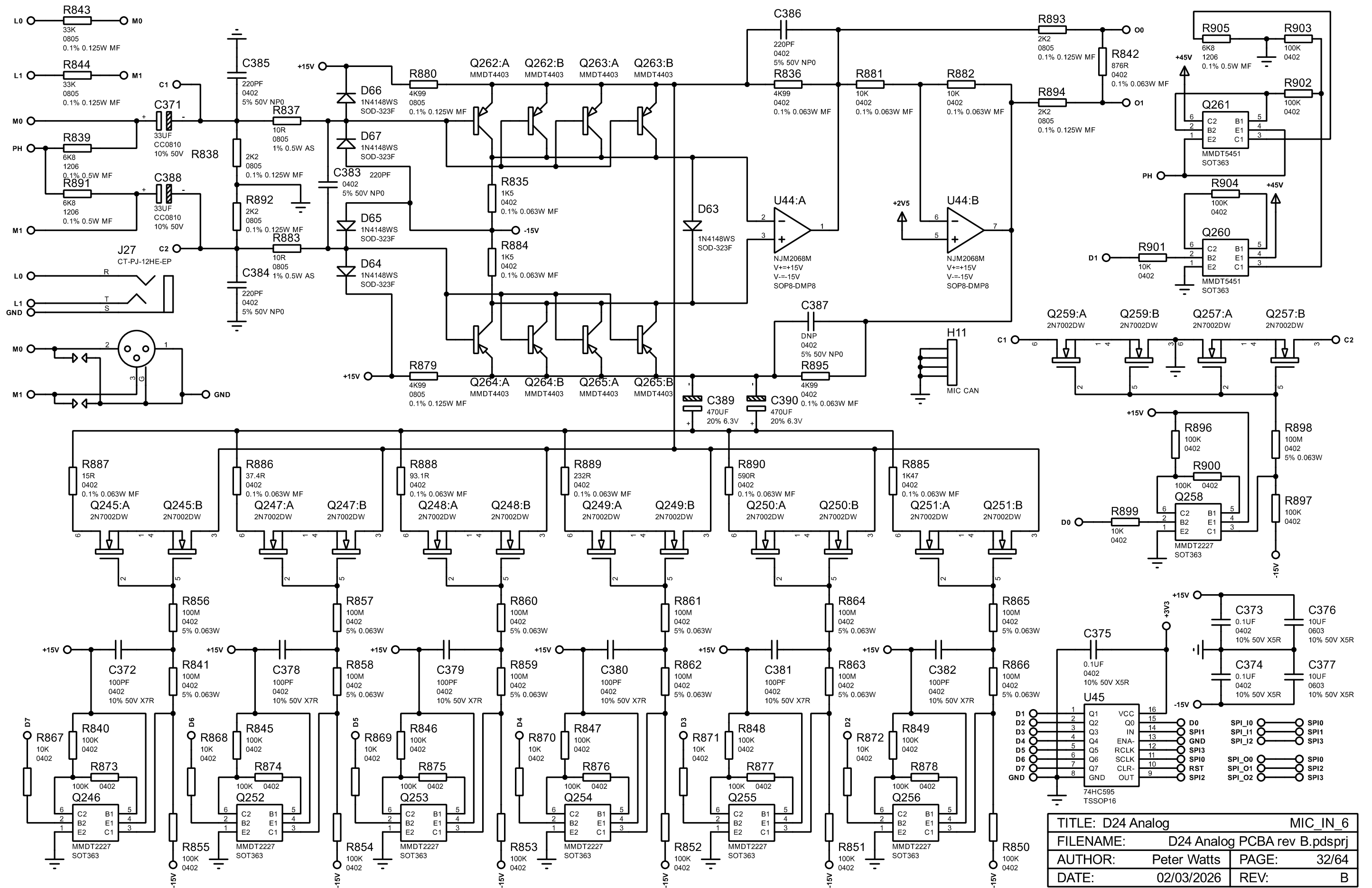
TITLE: D24 Analog		DC_15V
FILENAME: D24 Analog PCBA rev B.pdsprj		
AUTHOR: Peter Watts	PAGE: 29/64	
DATE: 02/03/2026	REV: B	



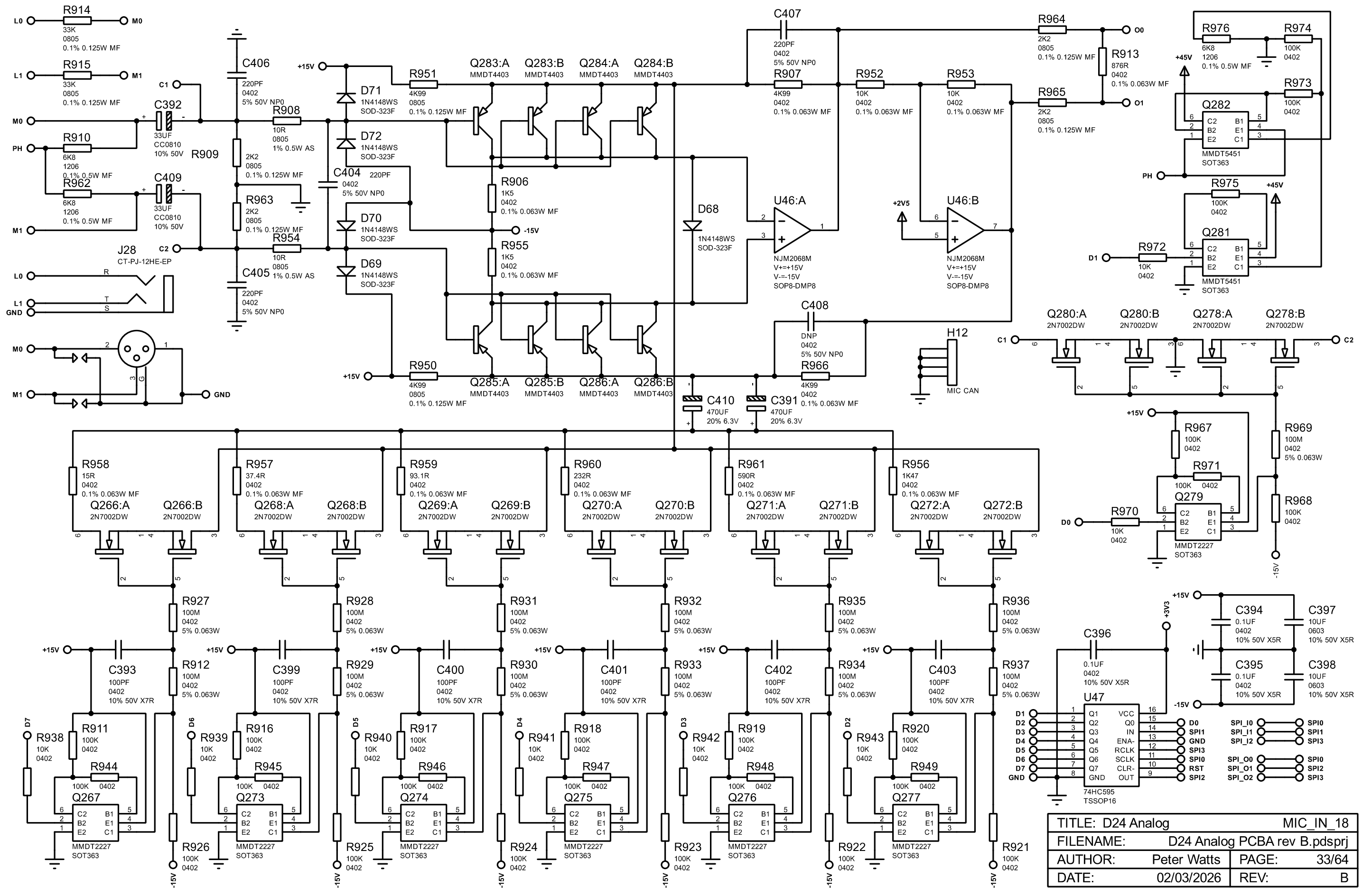
TITLE: D24 Analog		MIC_IN_5	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 30/64	
DATE: 02/03/2026		REV: B	



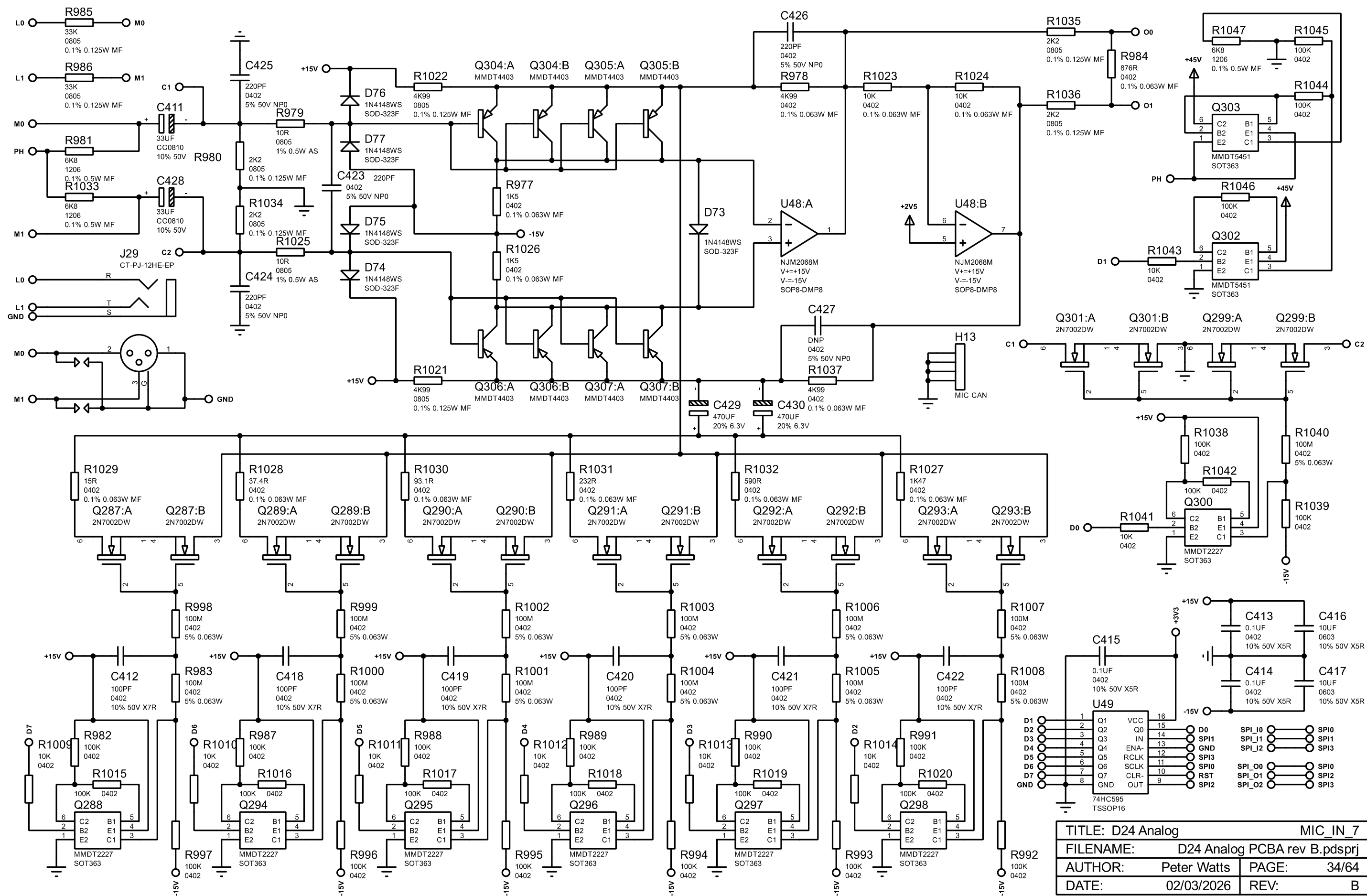
TITLE: D24 Analog		MIC_IN_17	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 31/64	
DATE: 02/03/2026		REV: B	

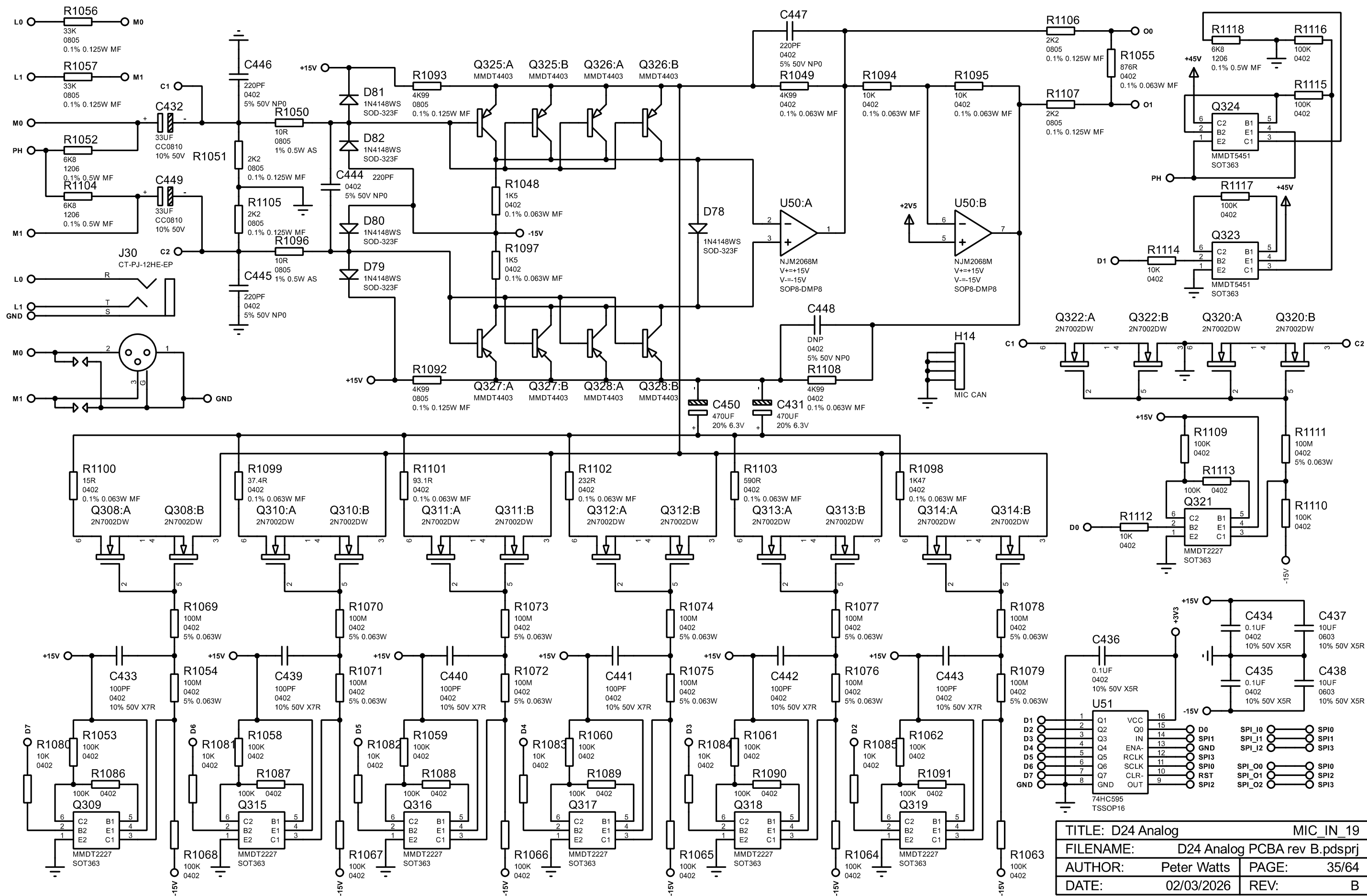


TITLE: D24 Analog		MIC_IN_6	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 32/64
DATE:		02/03/2026	REV: B

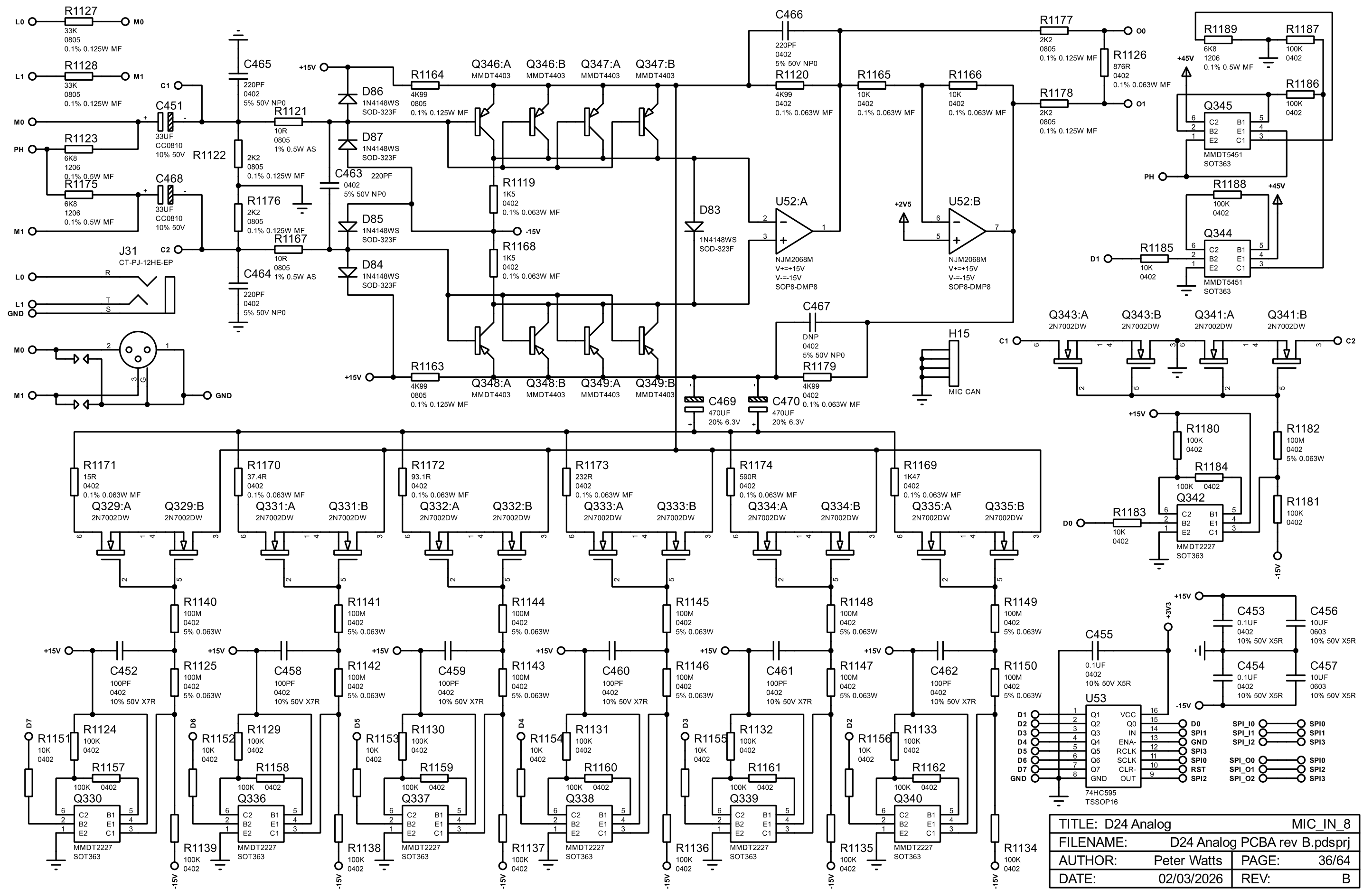


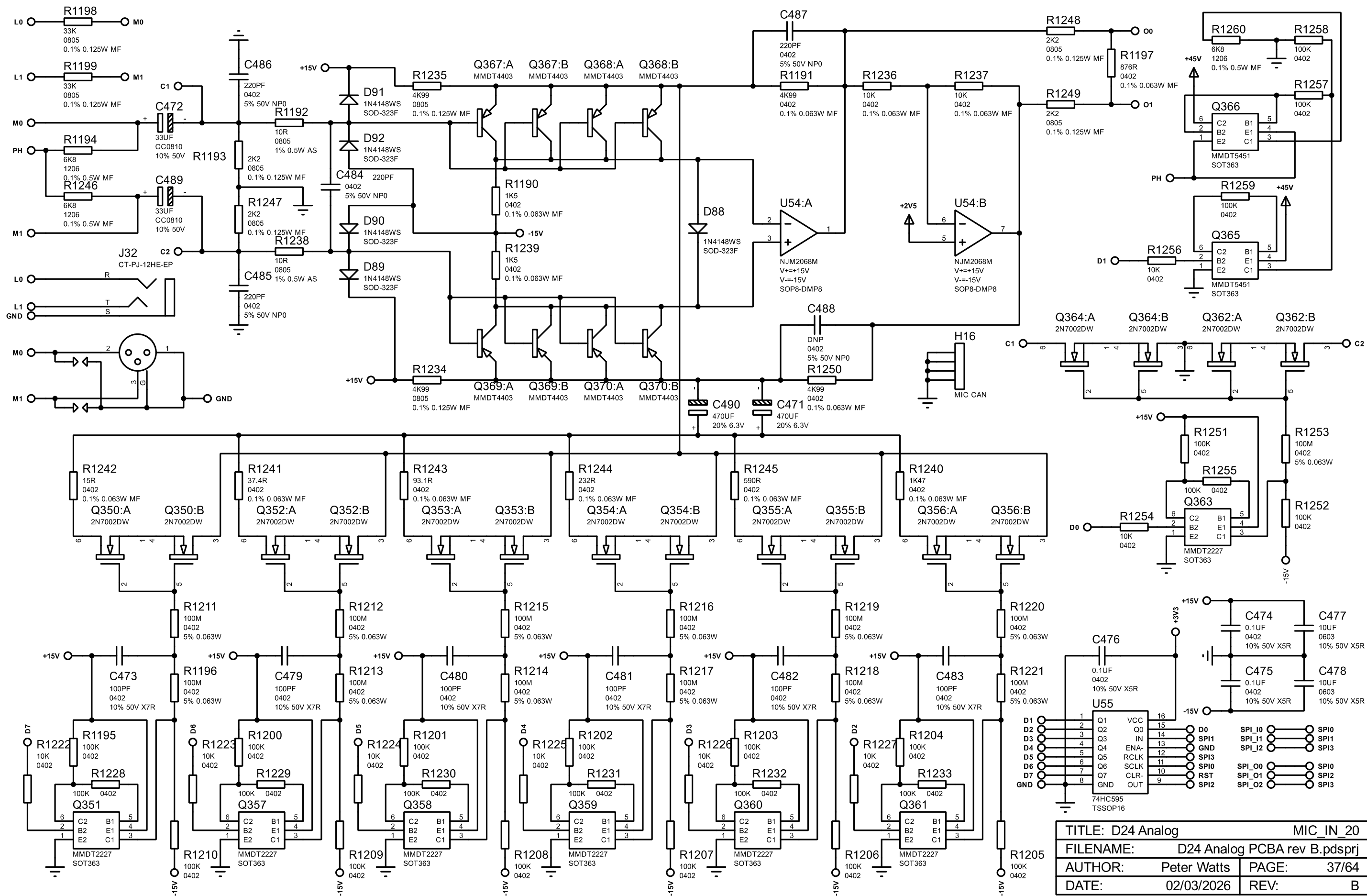
TITLE: D24 Analog		MIC_IN_18	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 33/64	
DATE: 02/03/2026		REV: B	





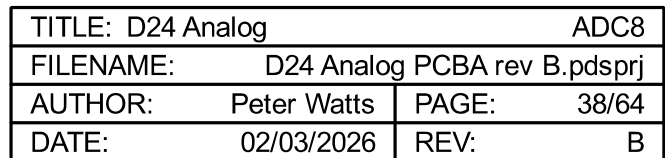
TITLE: D24 Analog		MIC_IN_19	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 35/64	
DATE: 02/03/2026		REV: B	

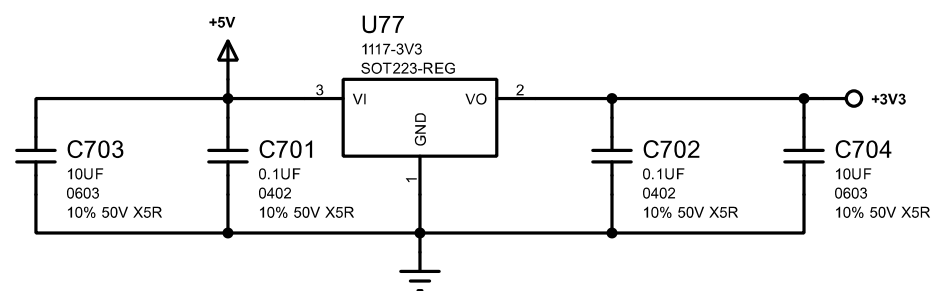




TITLE: D24 Analog		MIC_IN_20	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 37/64	
DATE: 02/03/2026		REV: B	

PINS:
TEST = 0 = internal 100k pull-down.
MCLK = 12.288MHz.
PW0-2 = 001 = Channel 1-8 output enable.
MSN = 0 = Slave mode.
CKS0-3 = 0100 = 48kHz, 12.288MHz, 256fs.
SLOW/SD = 01 = HPF short delay, sharp roll-off.
DIF0-1 = 10 = I2S 24-bit mode.
TDM0-1 = 01 = I2S 24-bit mode.
PSN = 1 = Parallel control mode.
I2C = 1 = Parallel control mode.
DP = 0 = PCM mode.
HPFE = 1 = High pass filter enable.
LDOE = 1 = Enable LDO voltages.
ODP = 1 = Normal TDM128 outputs.



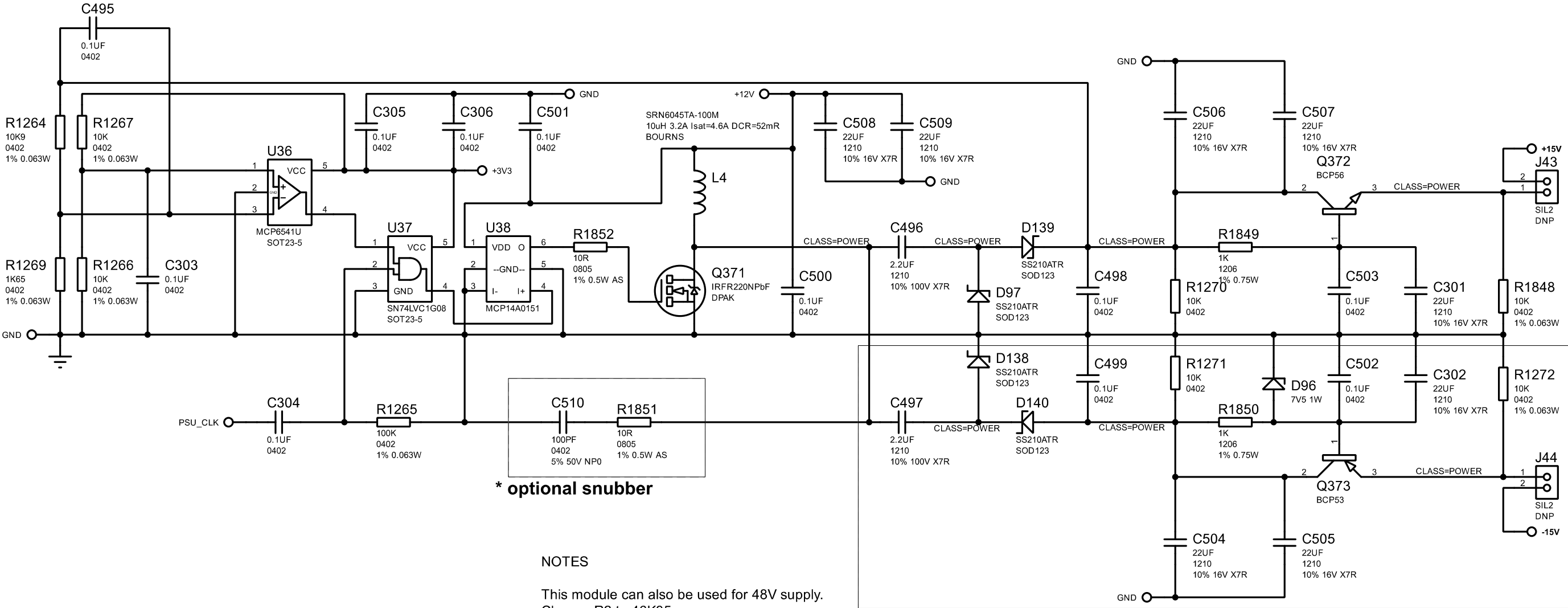


TITLE: D24 Analog		DC_3V3	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 39/64
DATE:		02/03/2026	REV: B

NOTES

gate res = 10R 0.5W metal

snubber 10R 1/8w as? 100pF C0G/NP0



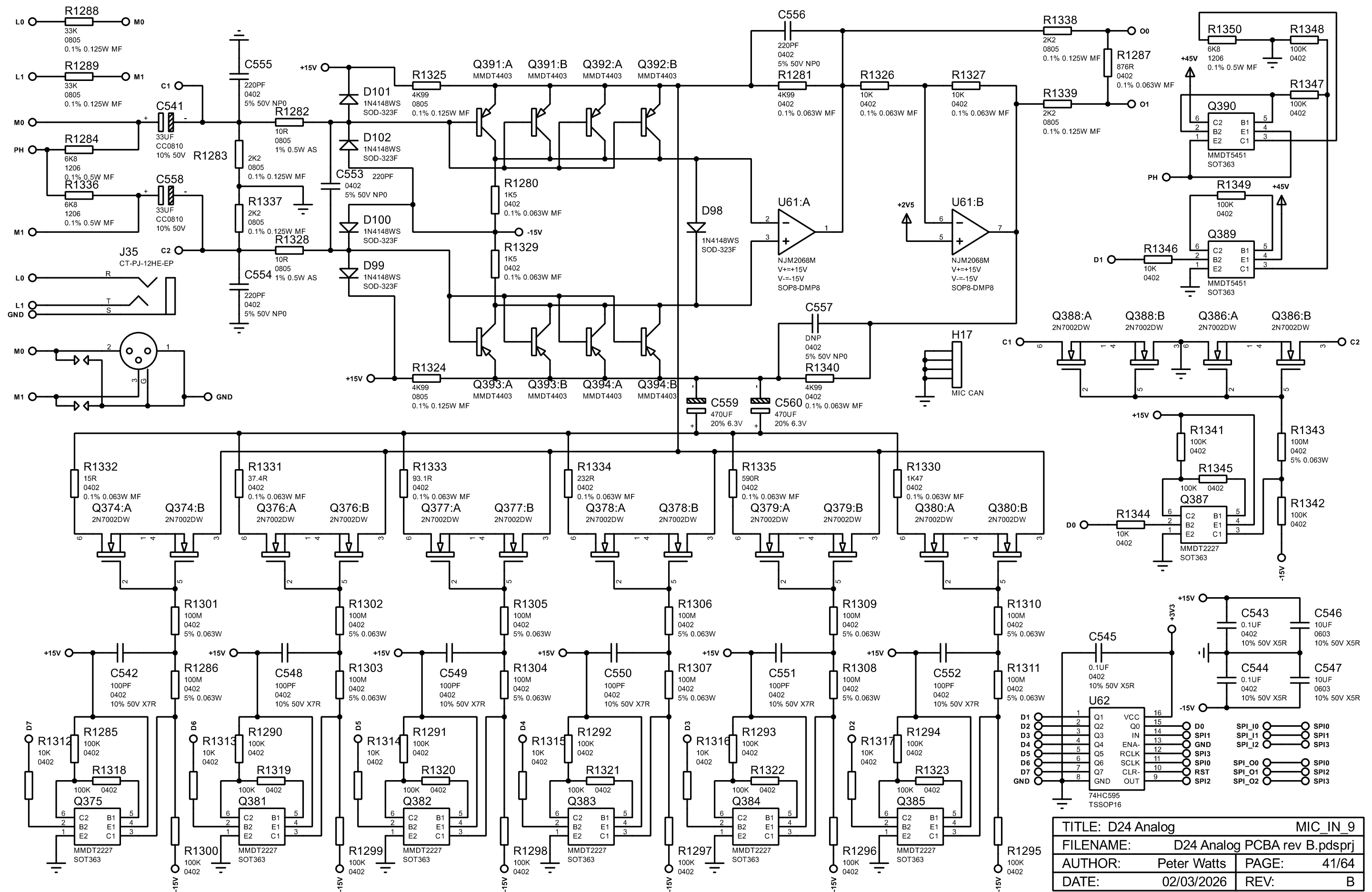
* optional snubber

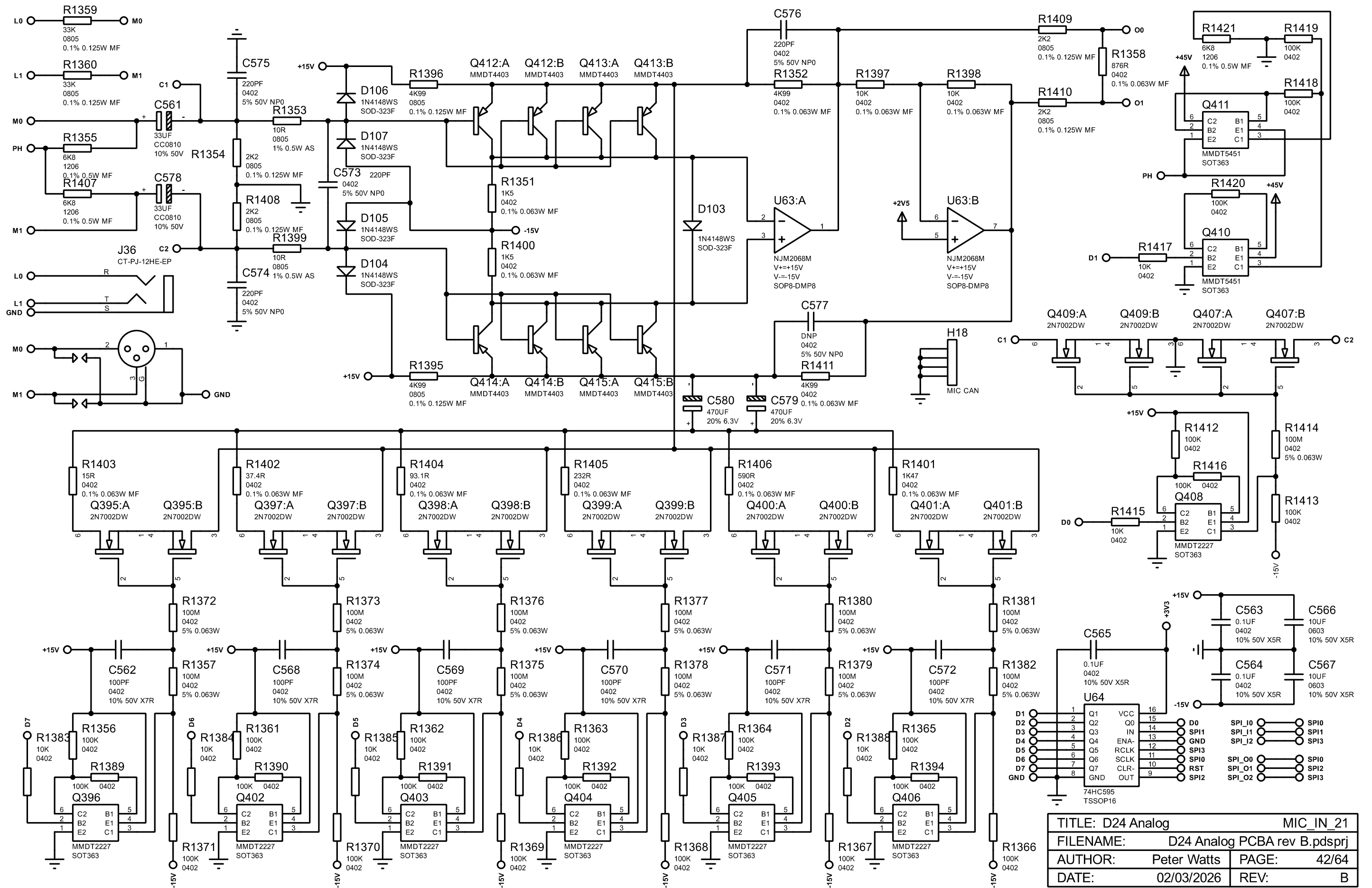
NOTES

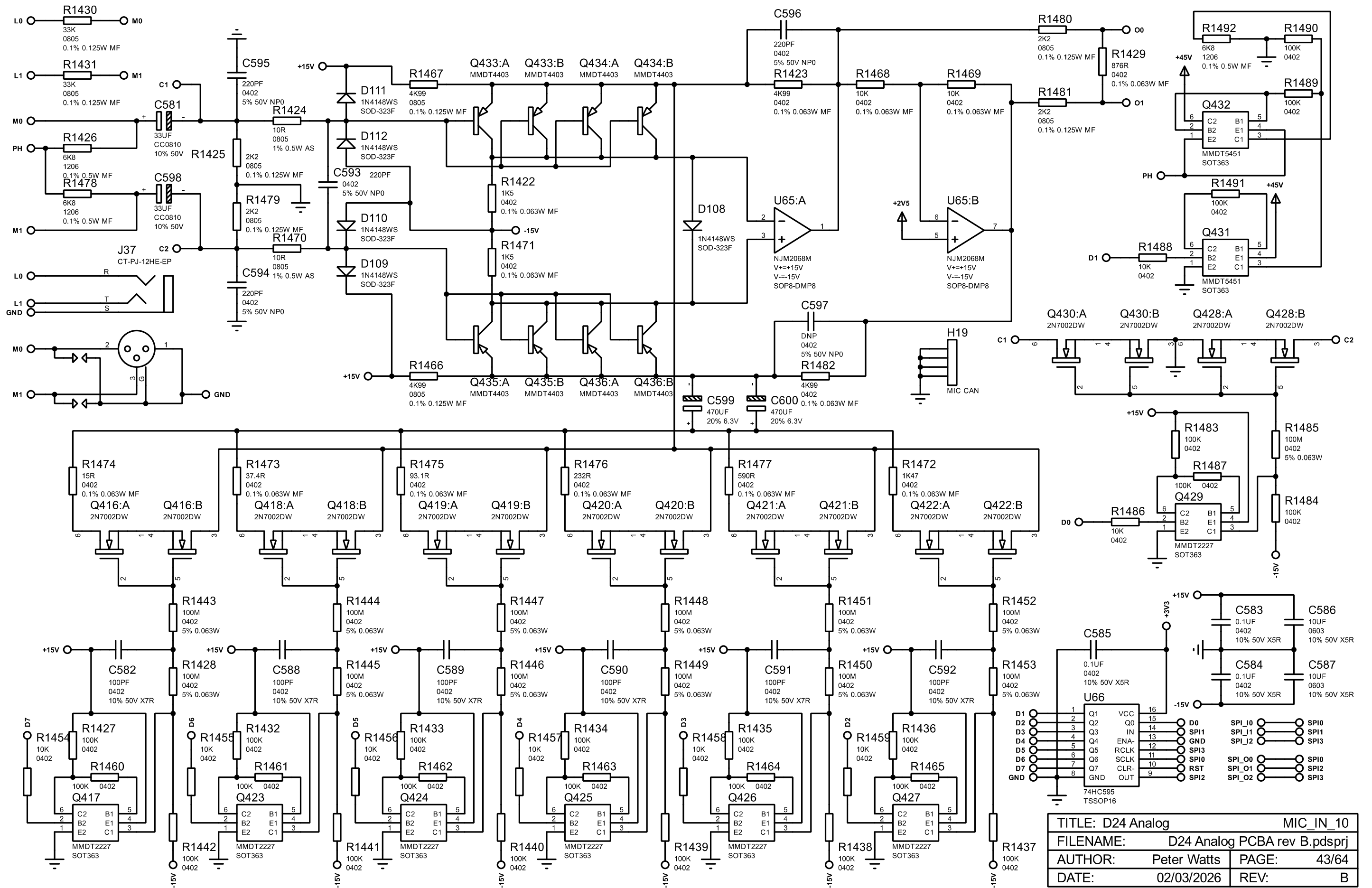
This module can also be used for 48V supply.
Change R2 to 46K35.
Change all 22UF caps to 10UF 50V.
Remove negative rail components.

* remove these components for 48V use

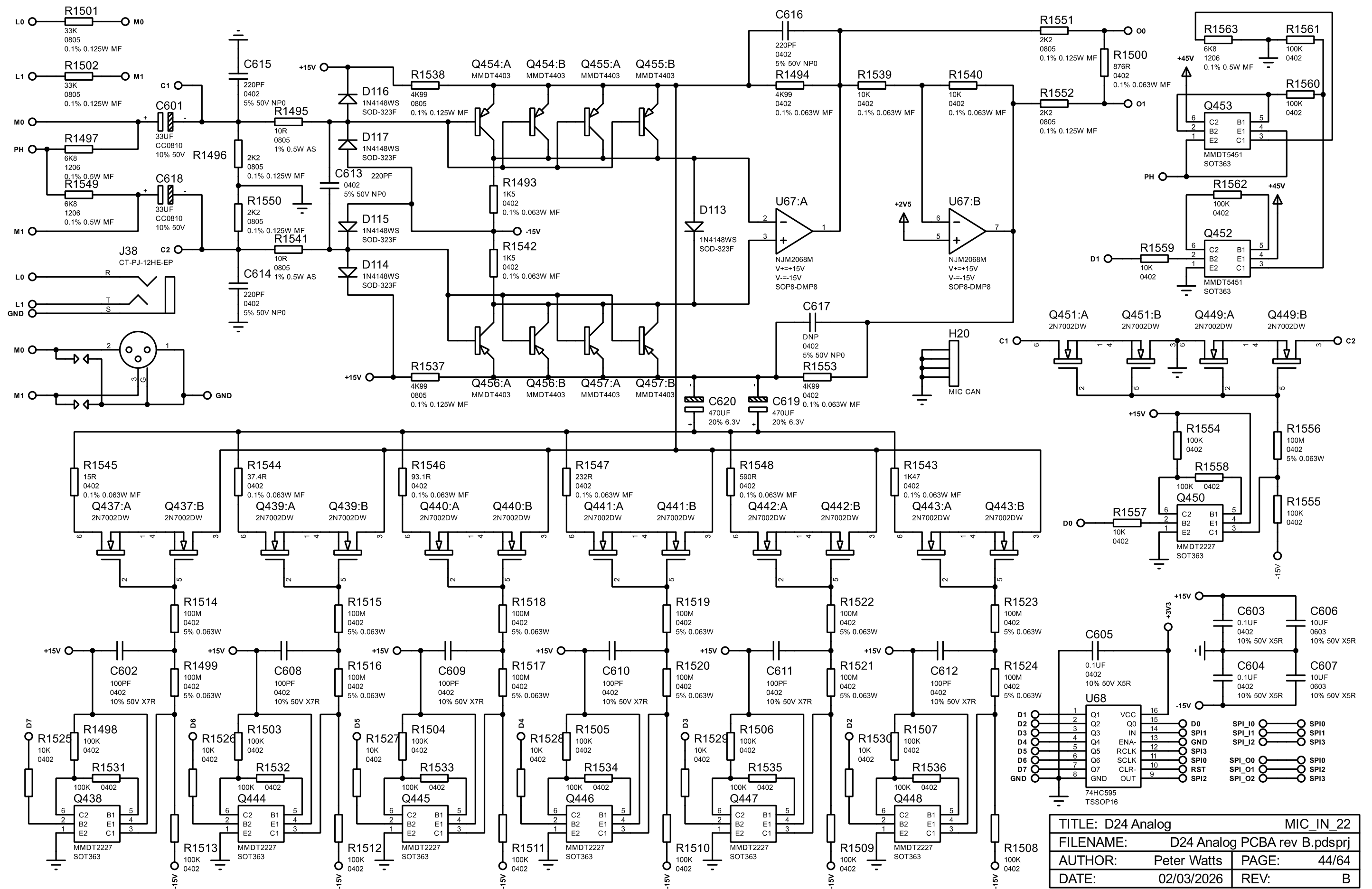
TITLE: D24 Analog		DC_15V	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 40/64
DATE:		02/03/2026	REV: B

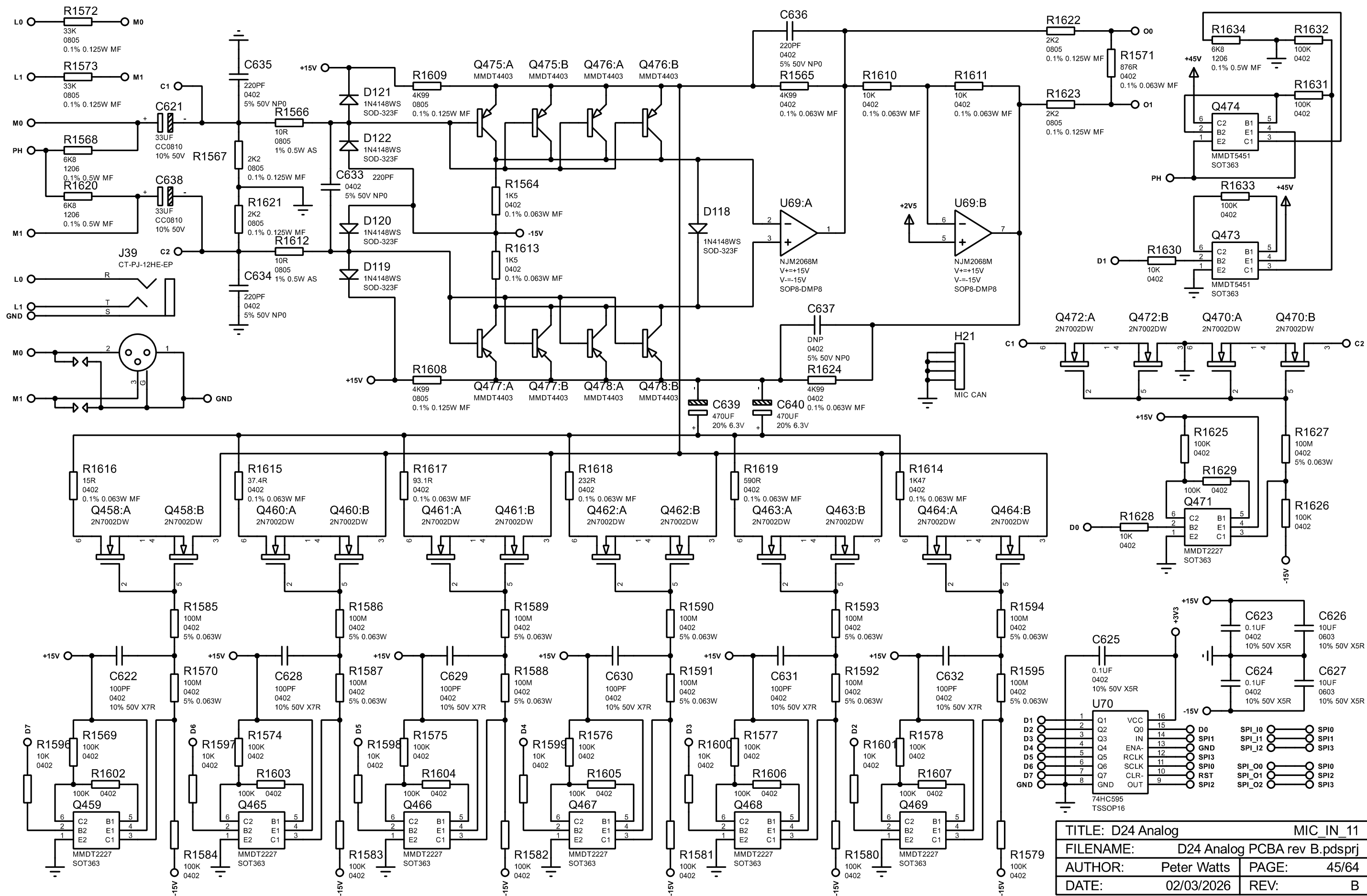




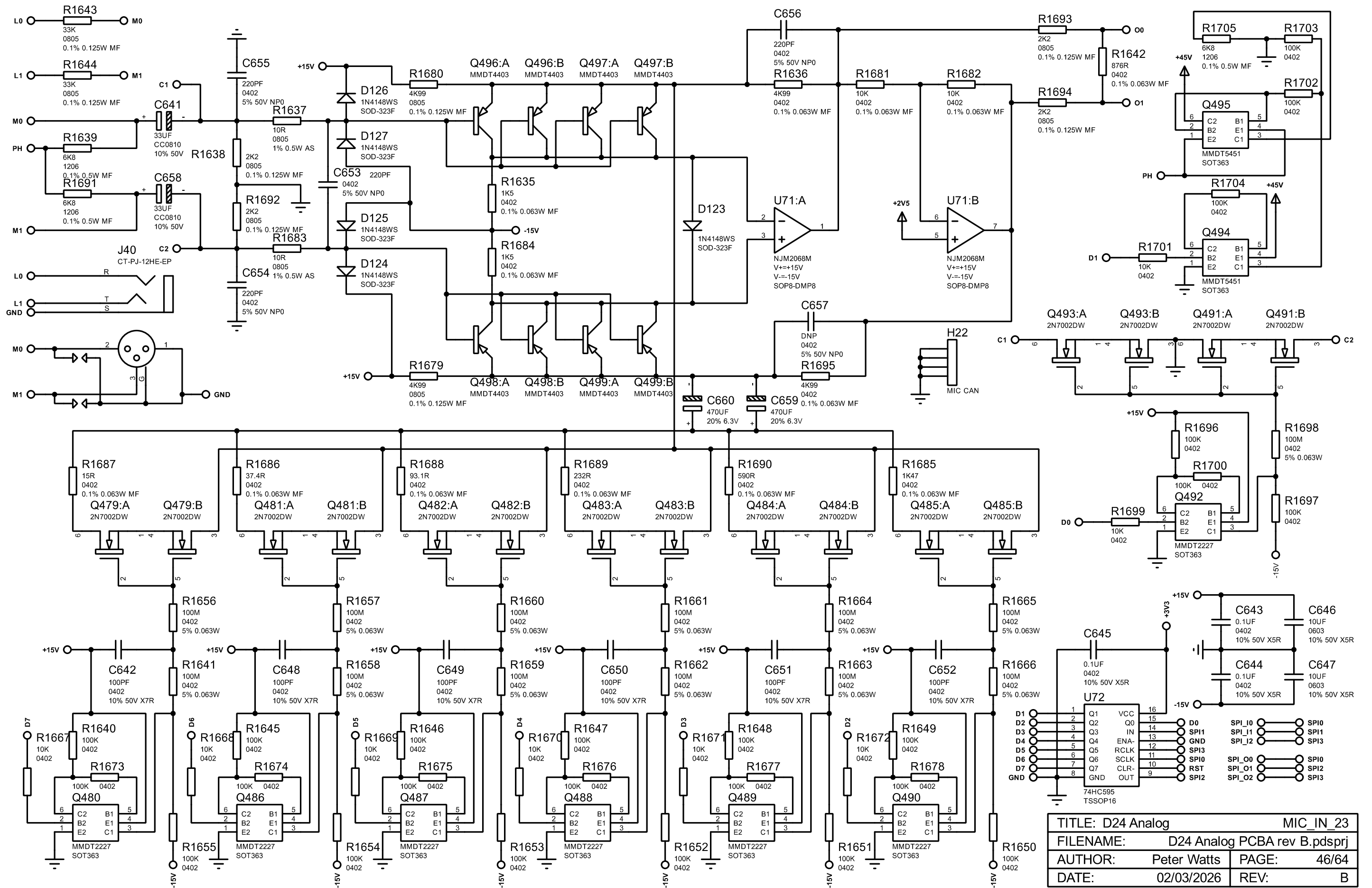


TITLE: D24 Analog		MIC_IN_10	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 43/64
DATE:		02/03/2026	REV: B

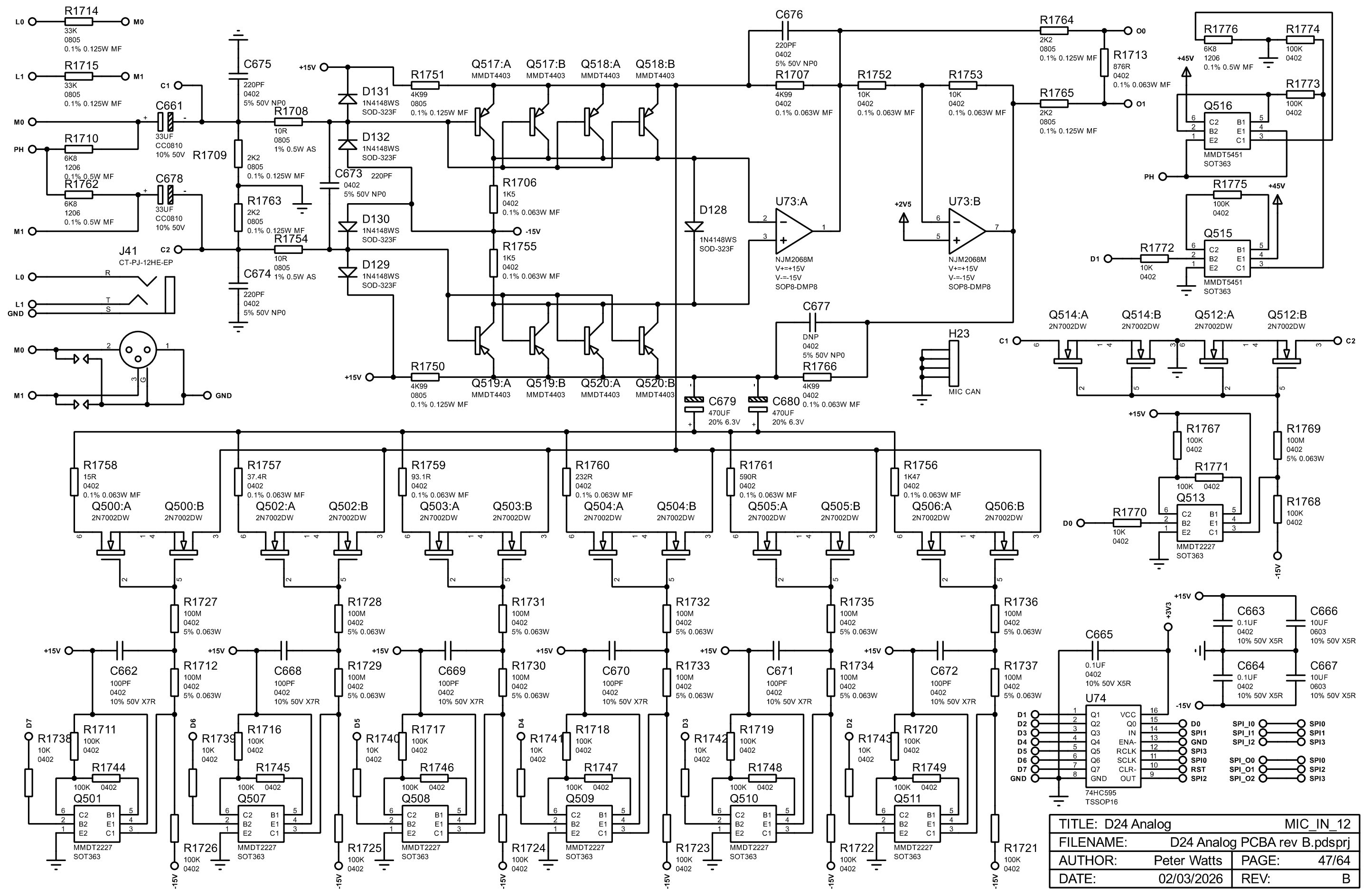


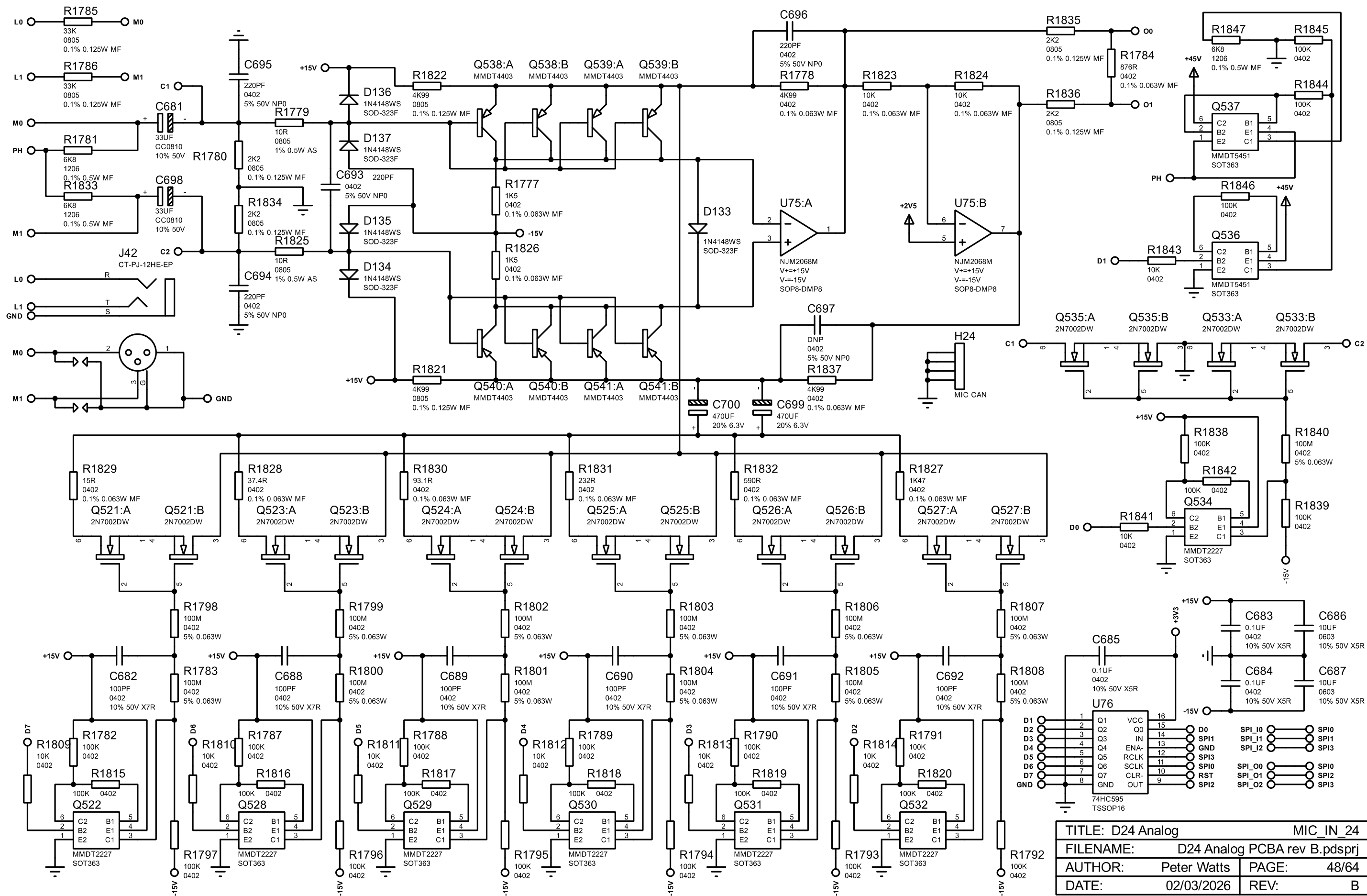


TITLE: D24 Analog		MIC_IN_11	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 45/64	
DATE: 02/03/2026		REV: B	



TITLE: D24 Analog		MIC_IN_23	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 46/64	
DATE: 02/03/2026		REV: B	





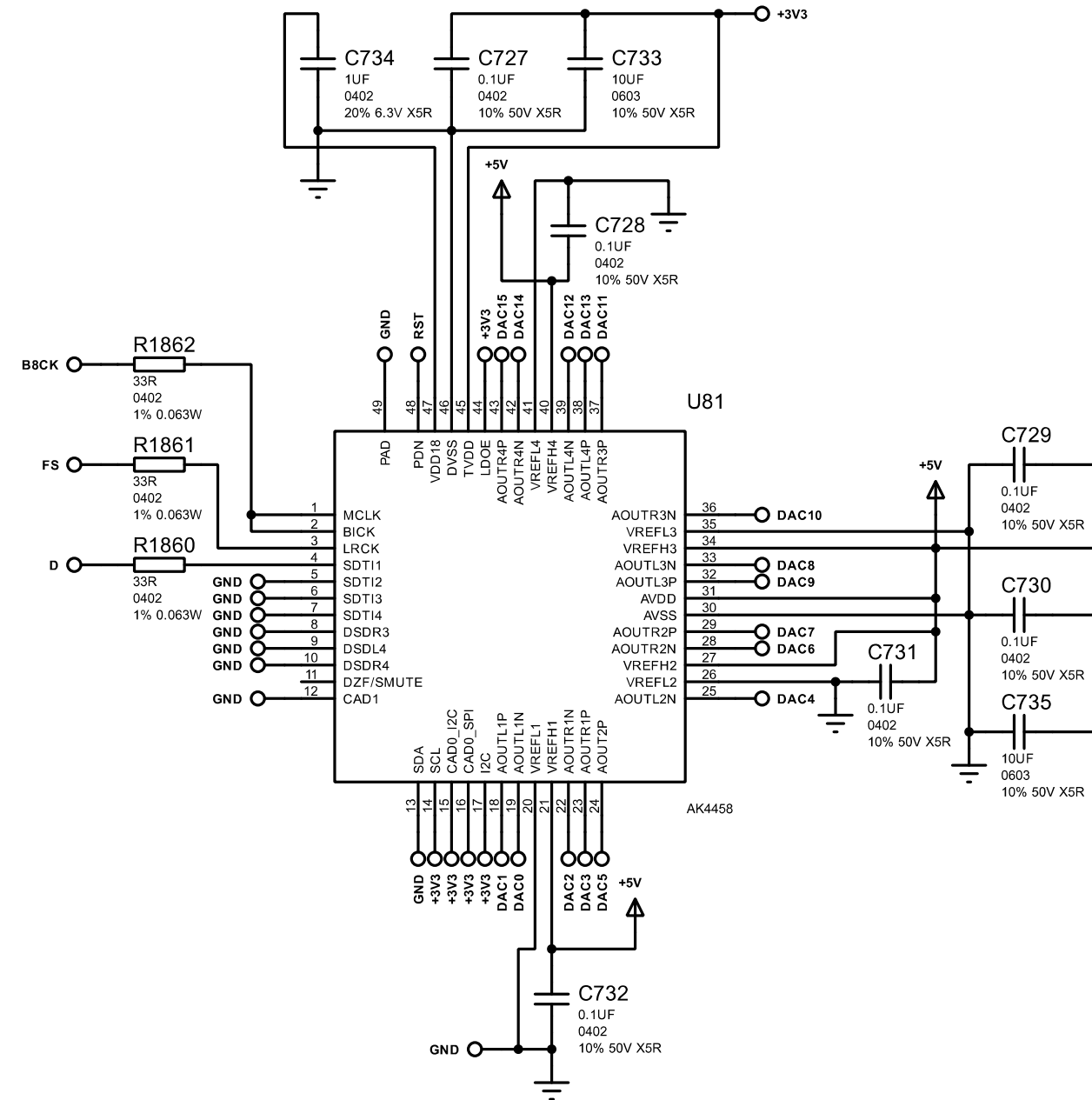
TITLE: D24 Analog		MIC_IN_24	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 48/64	
DATE: 02/03/2026		REV: B	

Place all 22R close to source.

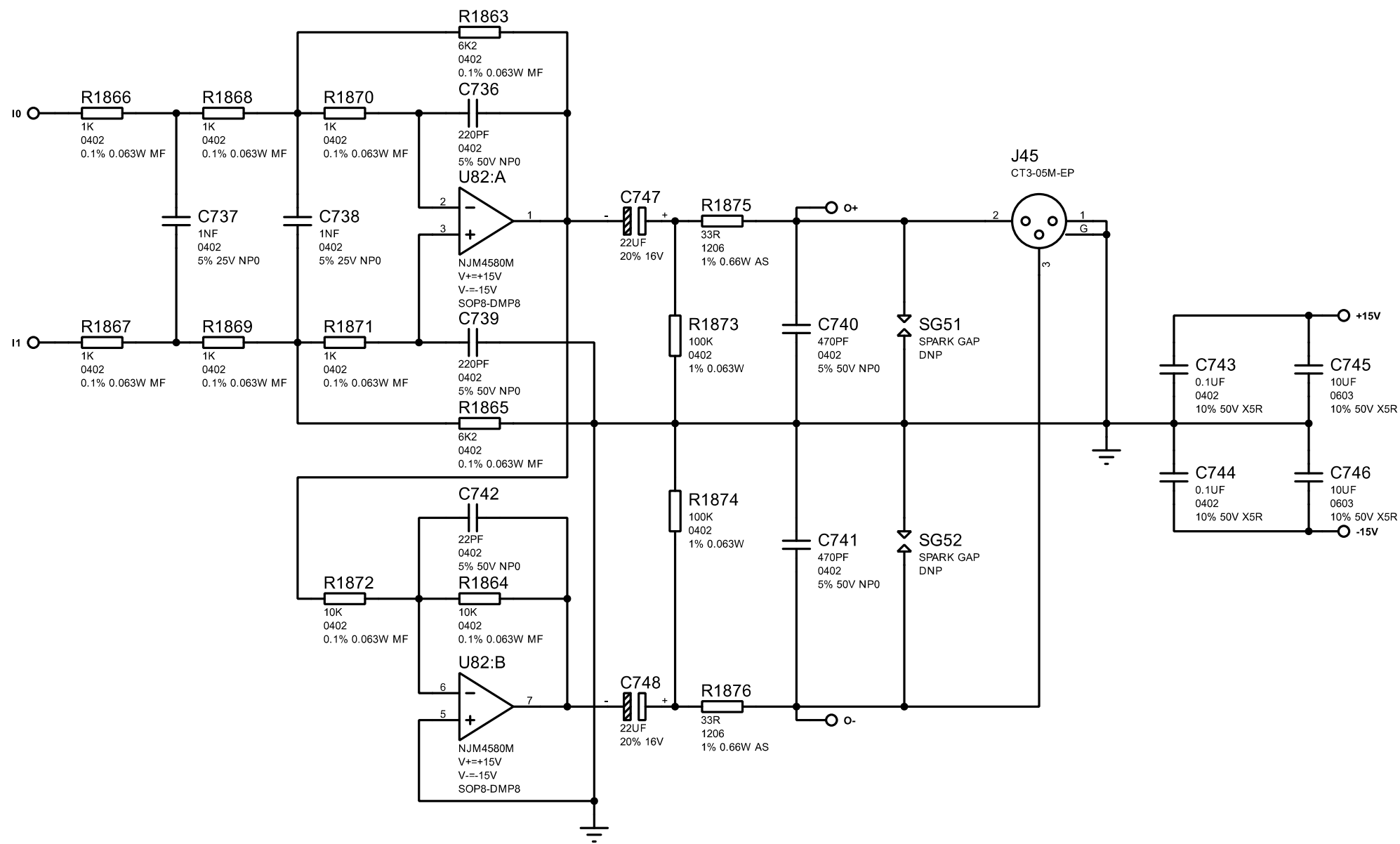
PINS:

CAD0_I2C = 1 = 32-bit I2S.
CAD0_SPI = 1 = Parallel control mode.
I2C = 1 = Parallel control mode.

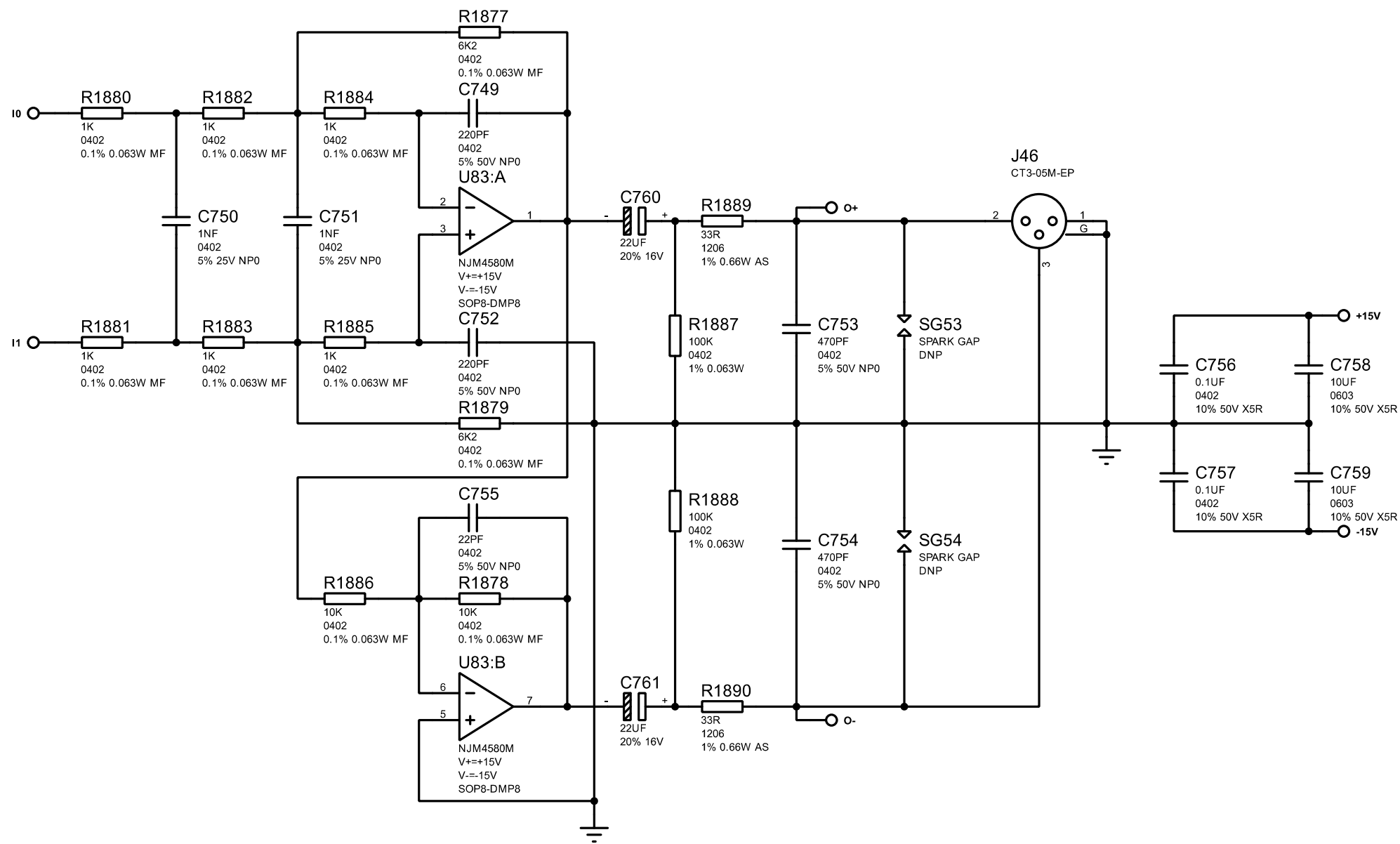
TEST = 0 = Internal 100k pull-down.
MCLK = 12.288MHz.
PW0-2 = 001 = Channel 1-8 output enable.
MSN = 0 = Slave mode.
CKS0-3 = 0100 = 48kHz, 12.288MHz, 256fs.
SLOW/SD = 01 = HPF short delay, sharp roll-off.
DIF0-1 = 10 = I2S 24-bit mode.
TDM0-1 = 01 = I2S 24-bit mode.
PSN = 1 = Parallel control mode.
I2C = 1 = Parallel control mode.
DP = 0 = PCM mode.
HPFE = 1 = High pass filter enable.
LDOE = 1 = Enable LDO voltages.
ODP = 1 = Normal TDM128 outputs.



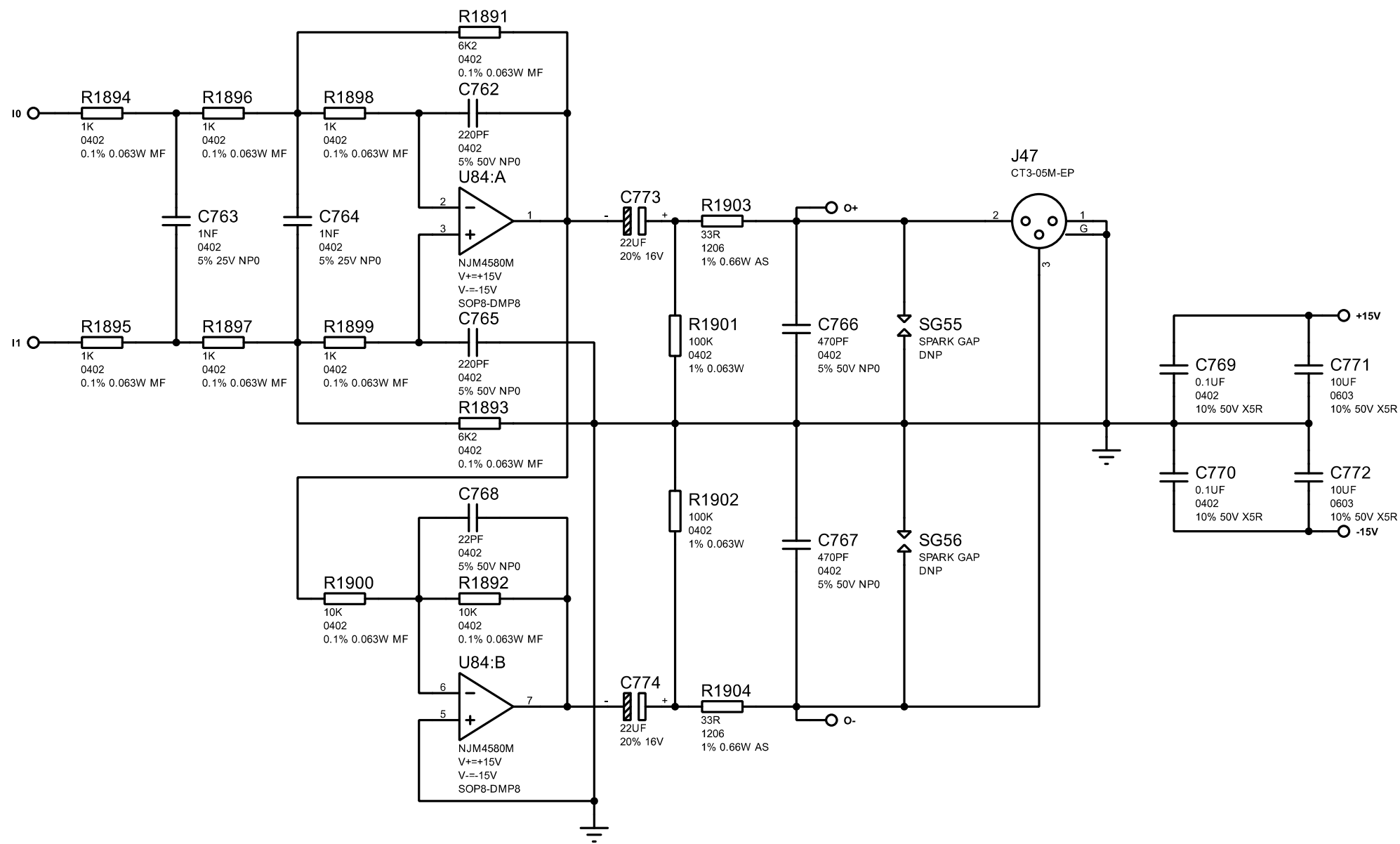
TITLE: D24 Analog		DAC8	
FILENAME: D24 Analog PCBA rev B.pdsprj			
AUTHOR: Peter Watts		PAGE: 49/64	
DATE: 02/03/2026		REV: B	



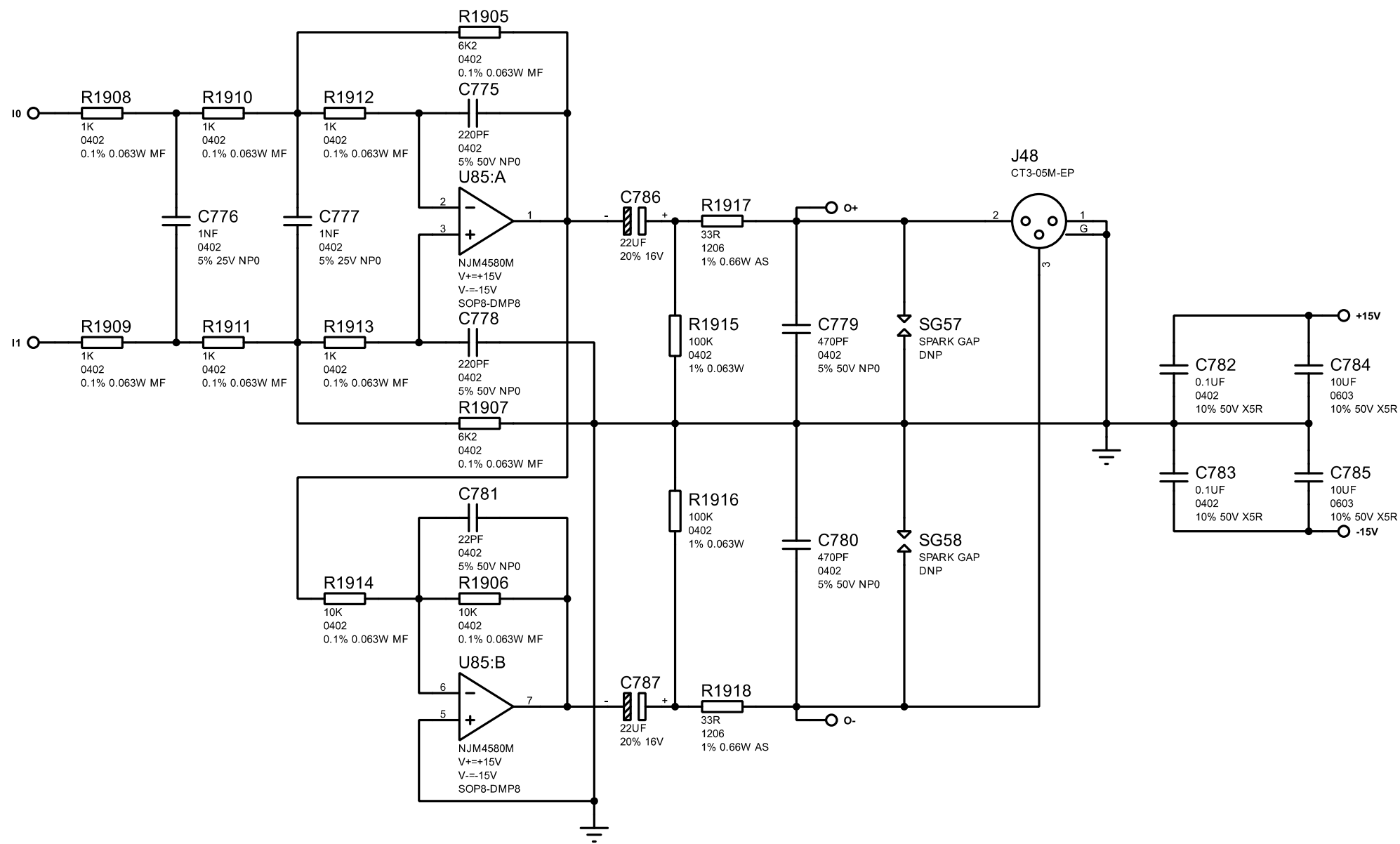
TITLE: D24 Analog		LIN_OUT_1	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 50/64
DATE:		02/03/2026	REV: B



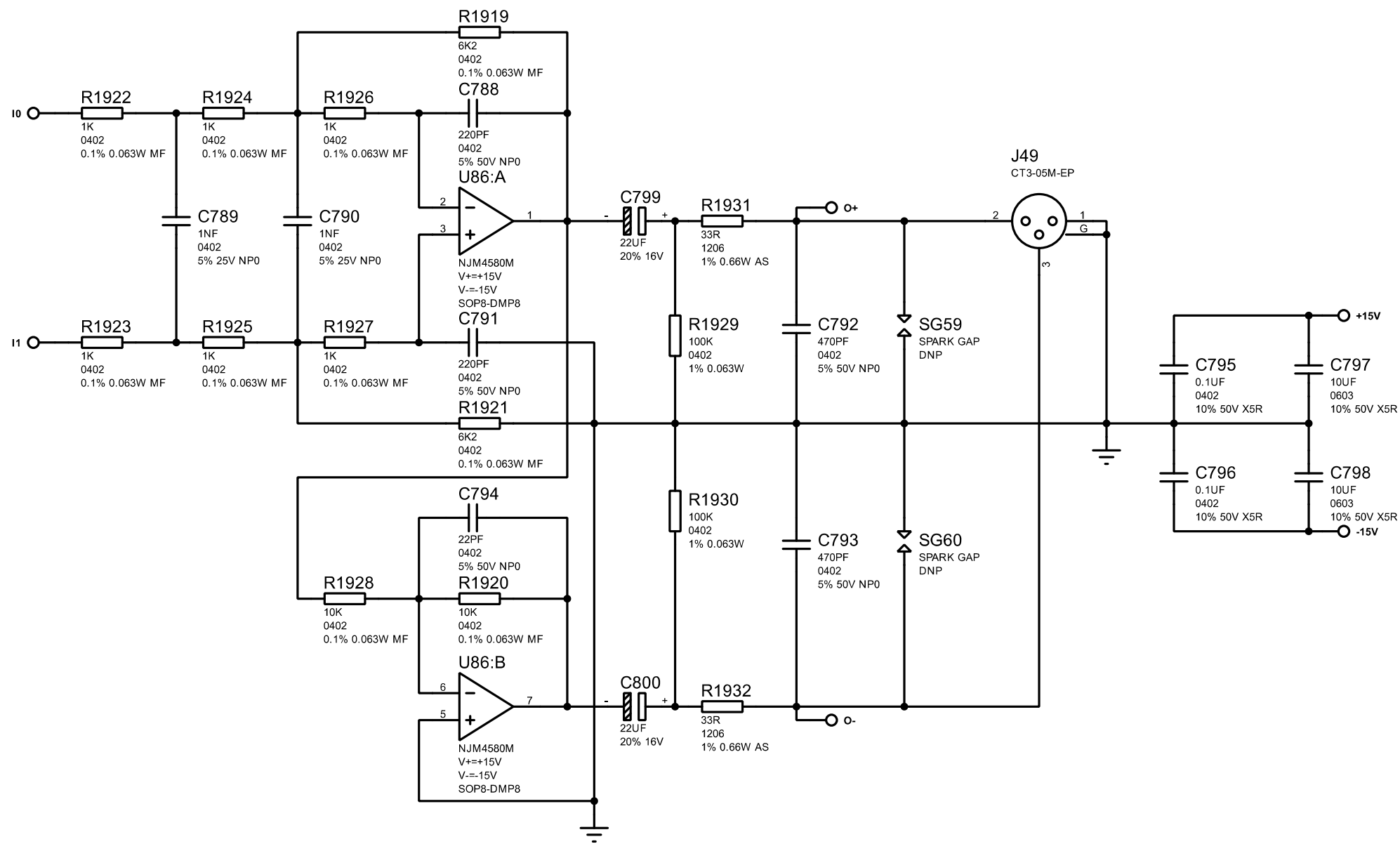
TITLE: D24 Analog		LIN_OUT_2	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 51/64
DATE:		02/03/2026	REV: B



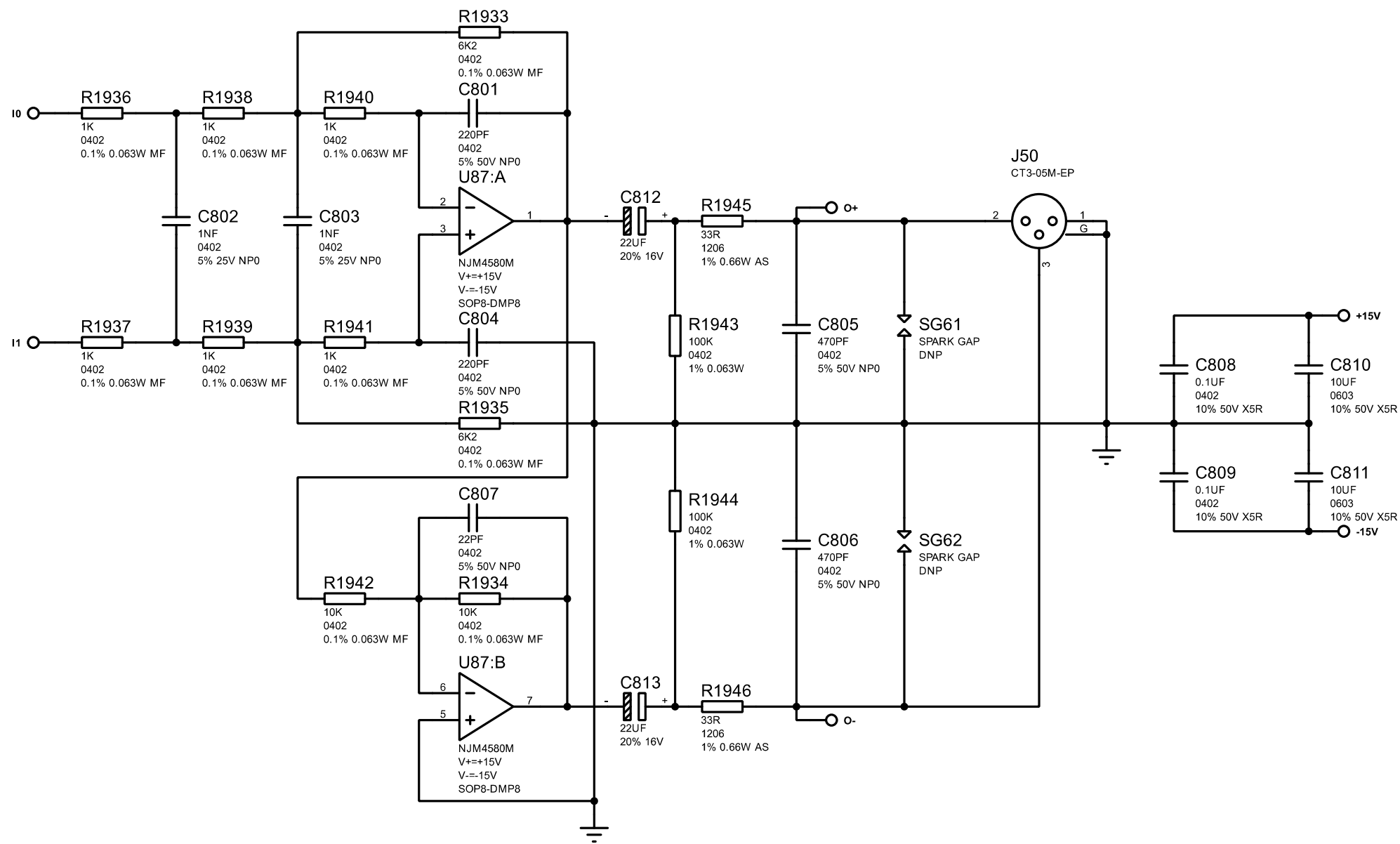
TITLE: D24 Analog		LIN_OUT_3	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 52/64
DATE:		02/03/2026	REV: B



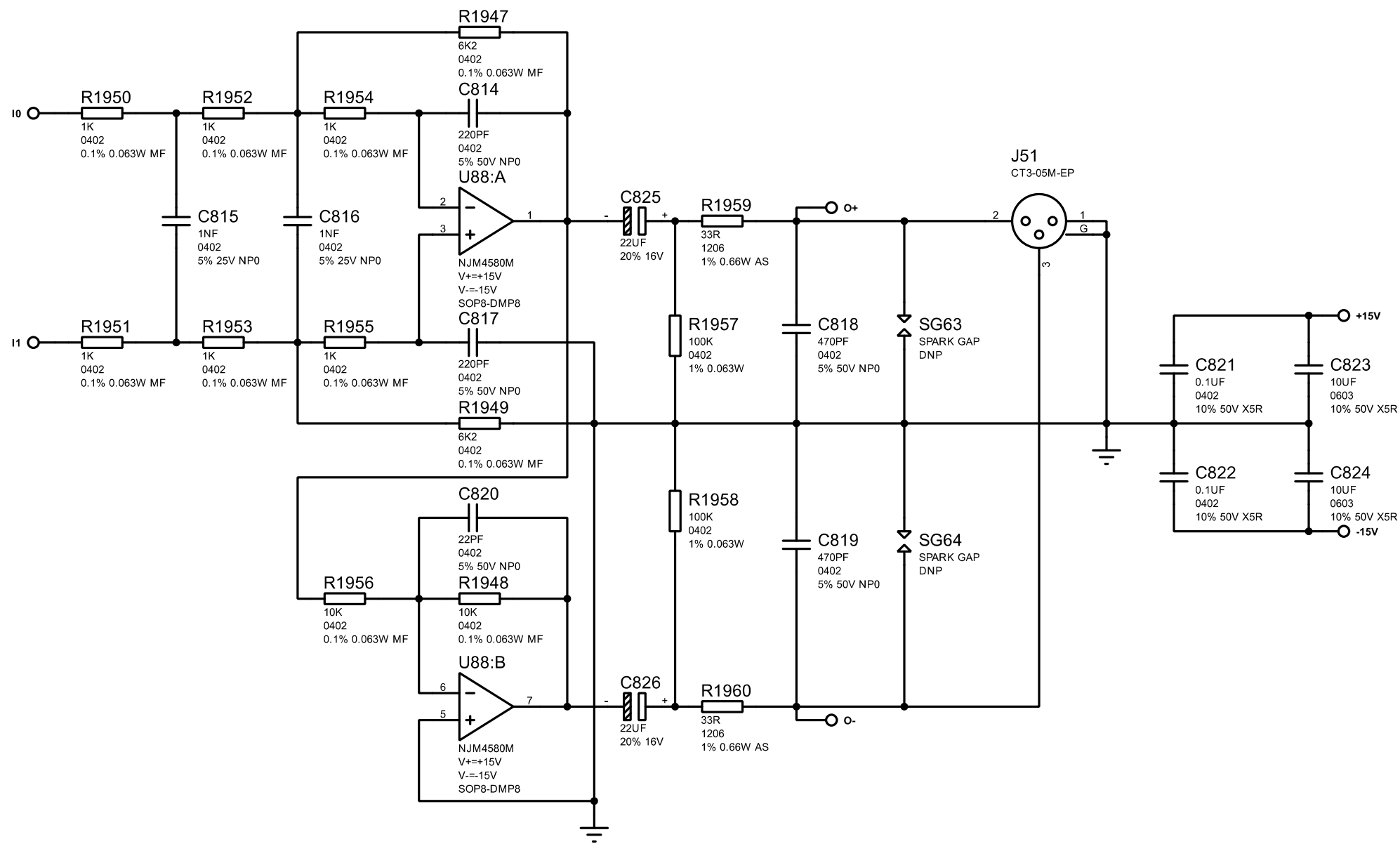
TITLE: D24 Analog		LIN_OUT_4	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 53/64
DATE:		02/03/2026	REV: B



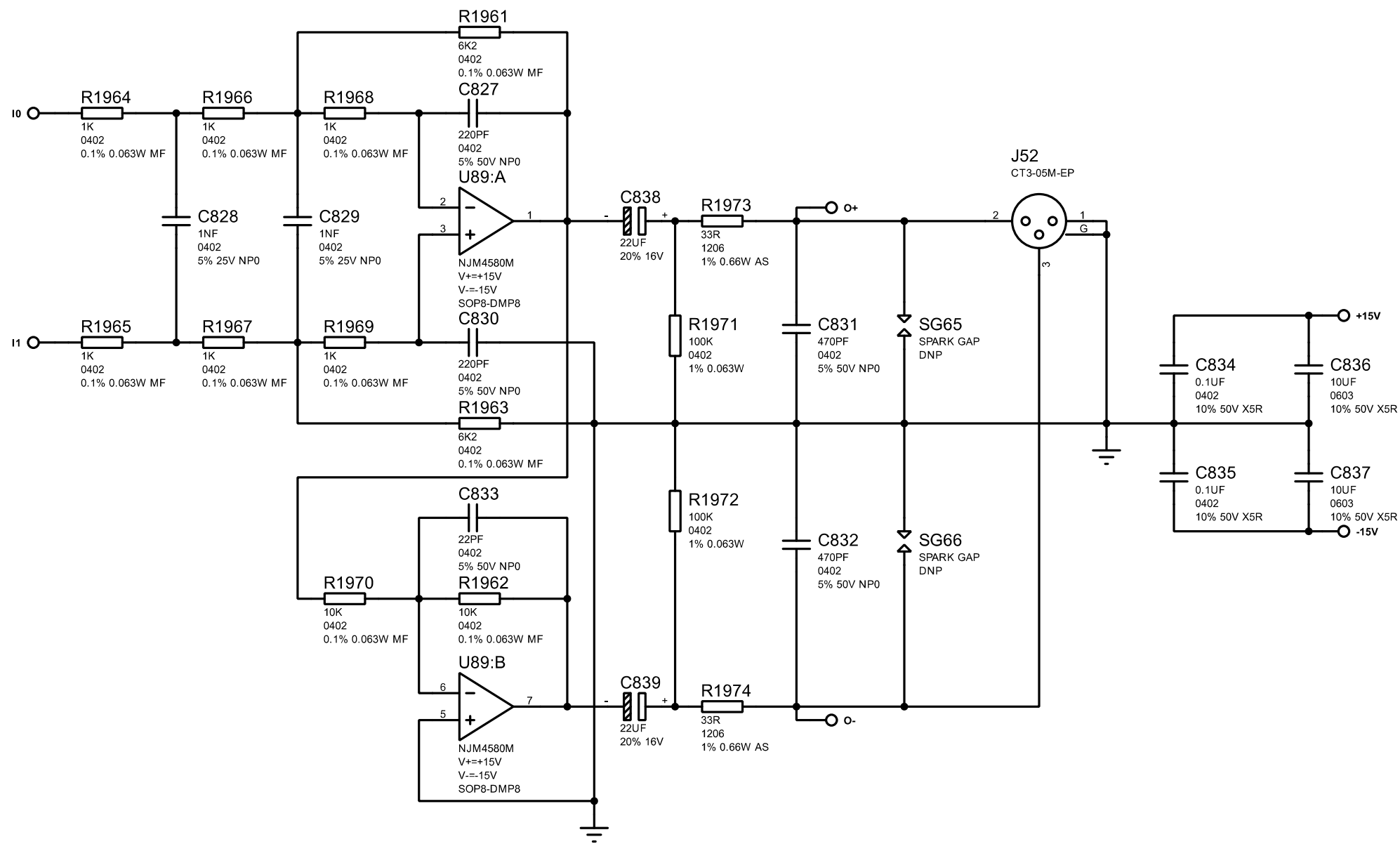
TITLE: D24 Analog		LIN_OUT_5	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 54/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		LIN_OUT_6	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 55/64
DATE:		02/03/2026	REV: B

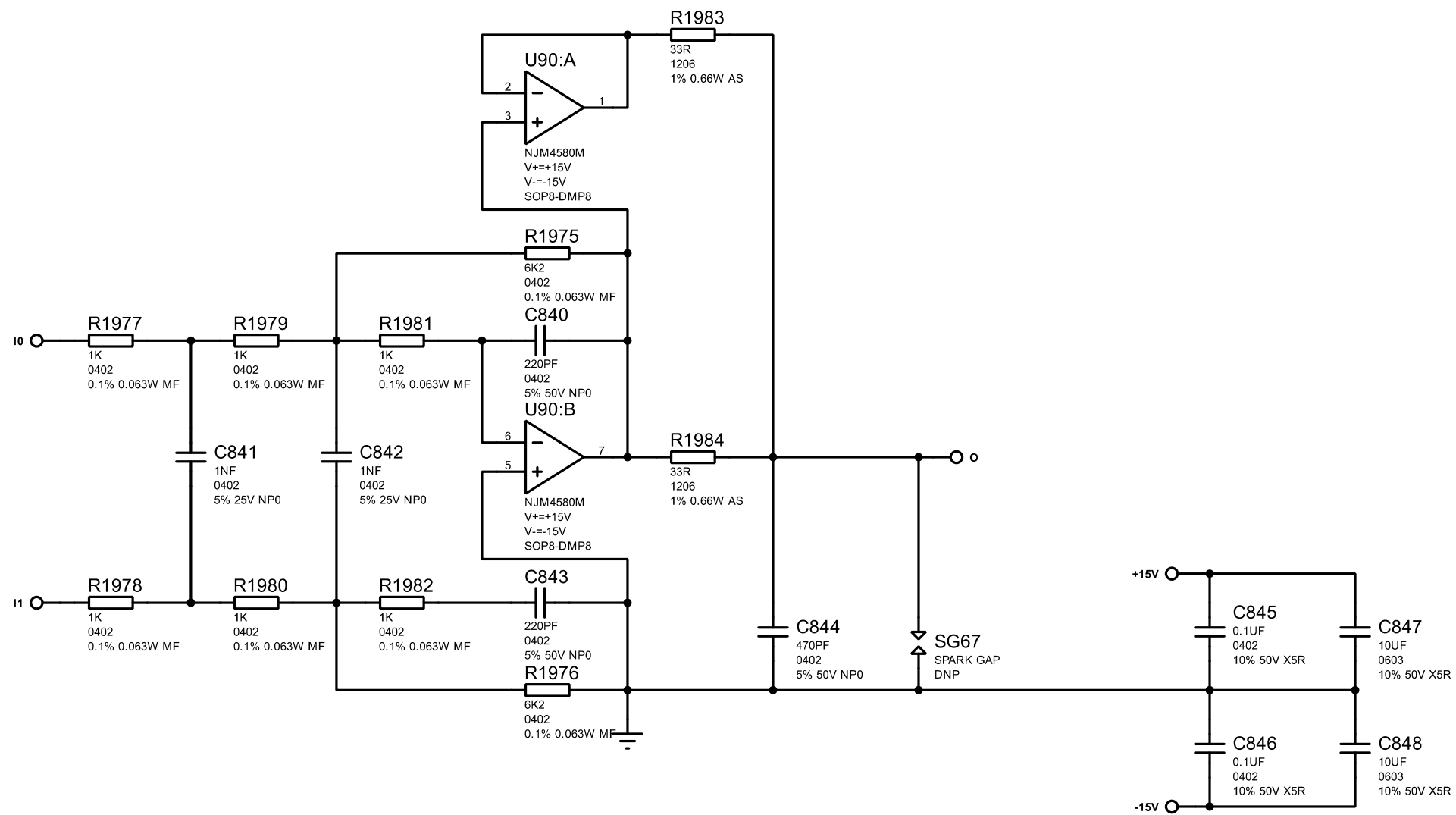


TITLE: D24 Analog		LIN_OUT_7	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 56/64
DATE:		02/03/2026	REV: B

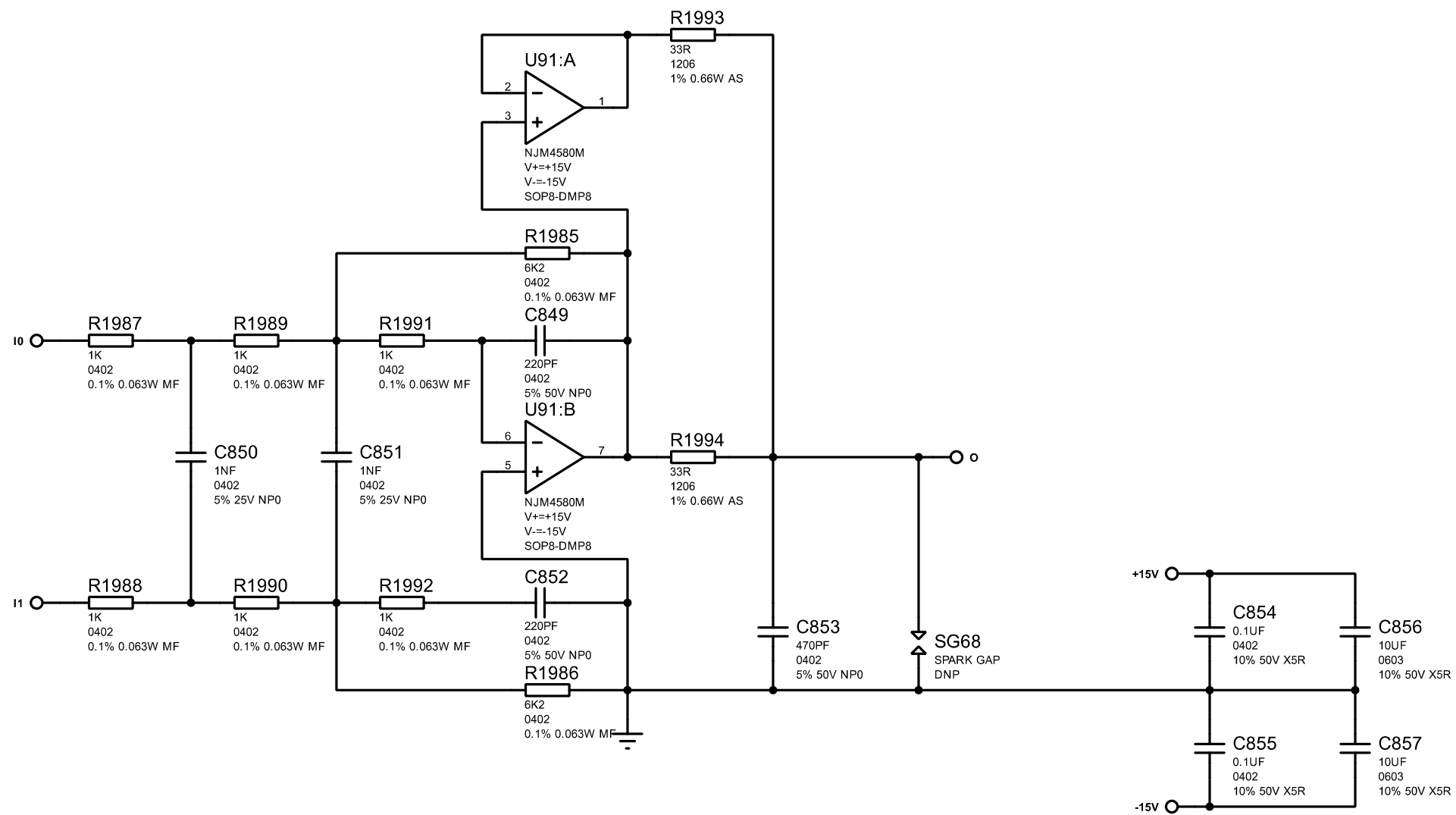


TITLE: D24 Analog		LIN_OUT_8	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 57/64
DATE:		02/03/2026	REV: B

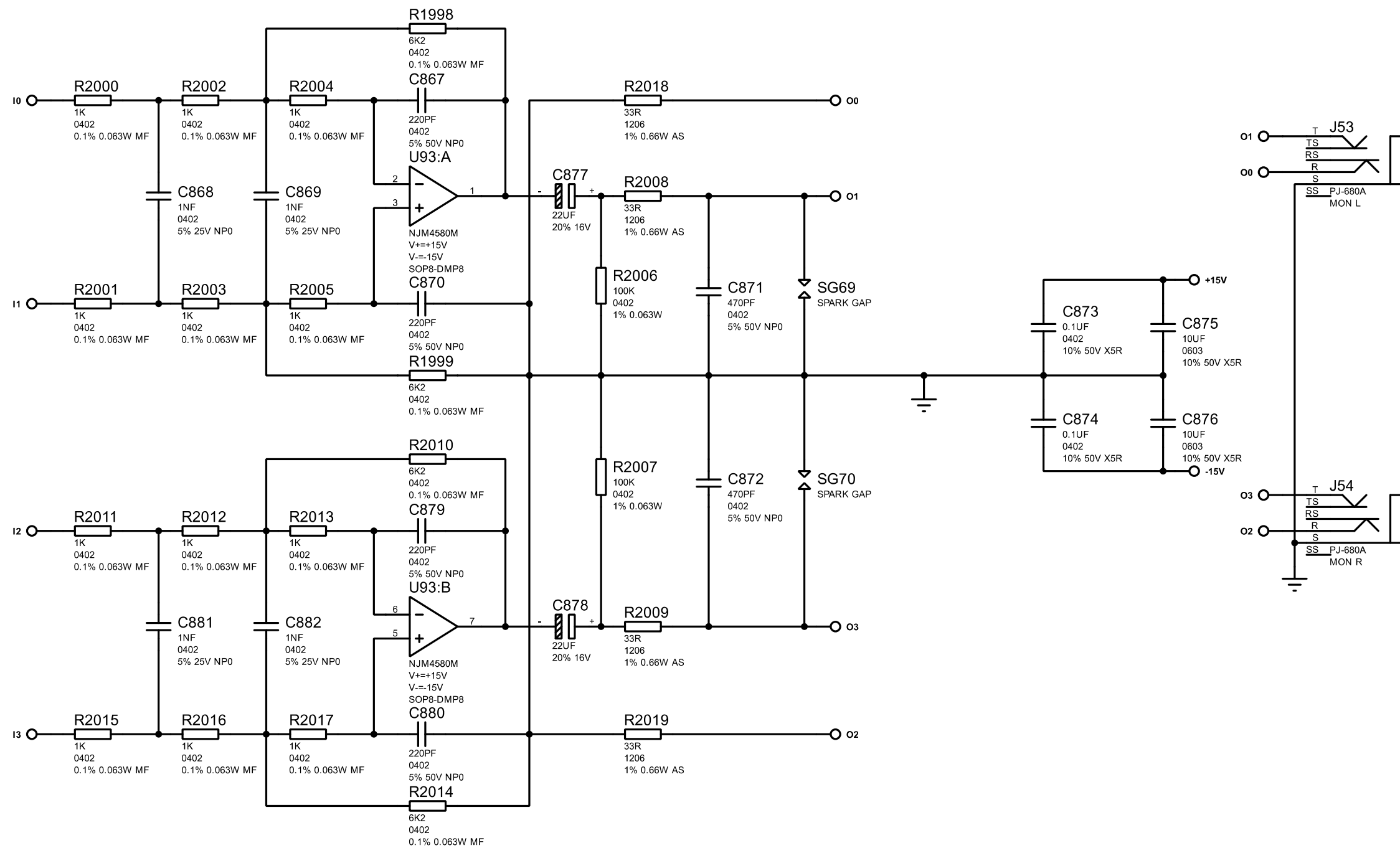
TITLE: D24 Analog		DAC8
FILENAME: D24 Analog PCBA rev B.pdsprj		
AUTHOR: Peter Watts	PAGE:	58/64
DATE: 02/03/2026	REV:	B



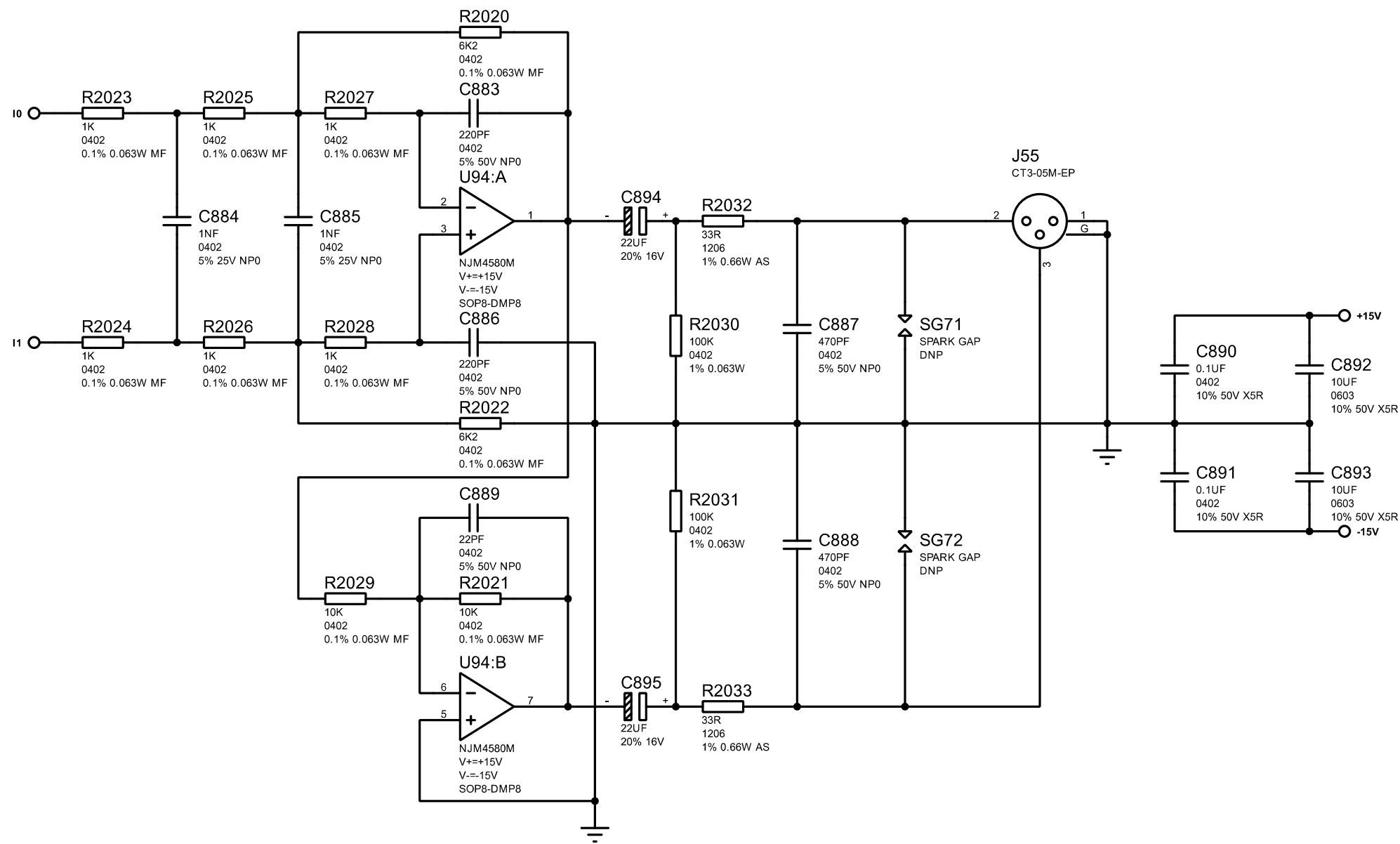
TITLE: D24 Analog		PHONES_L	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 59/64
DATE:		02/03/2026	REV: B



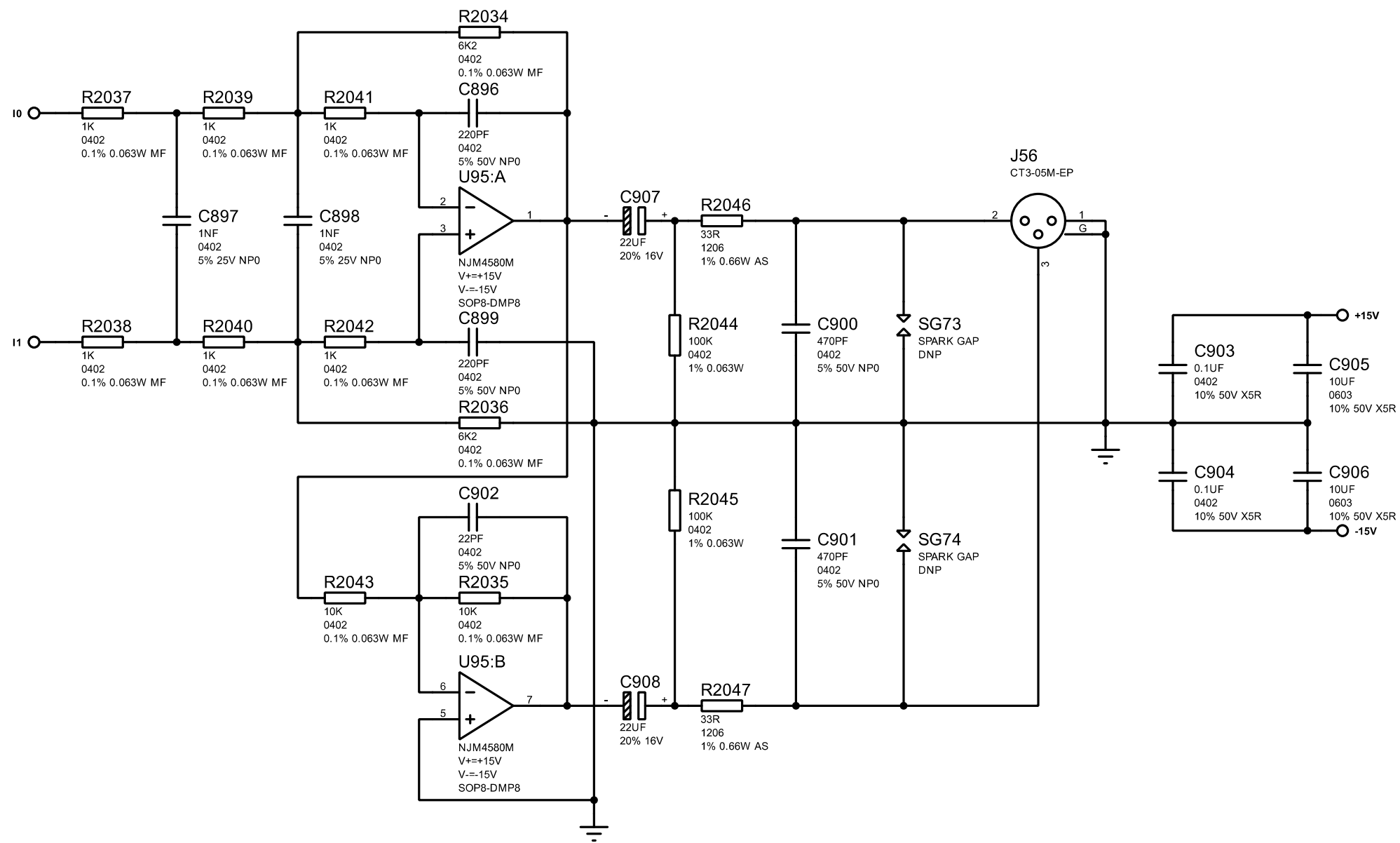
TITLE: D24 Analog		PHONES_R	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 60/64
DATE:		02/03/2026	REV: B



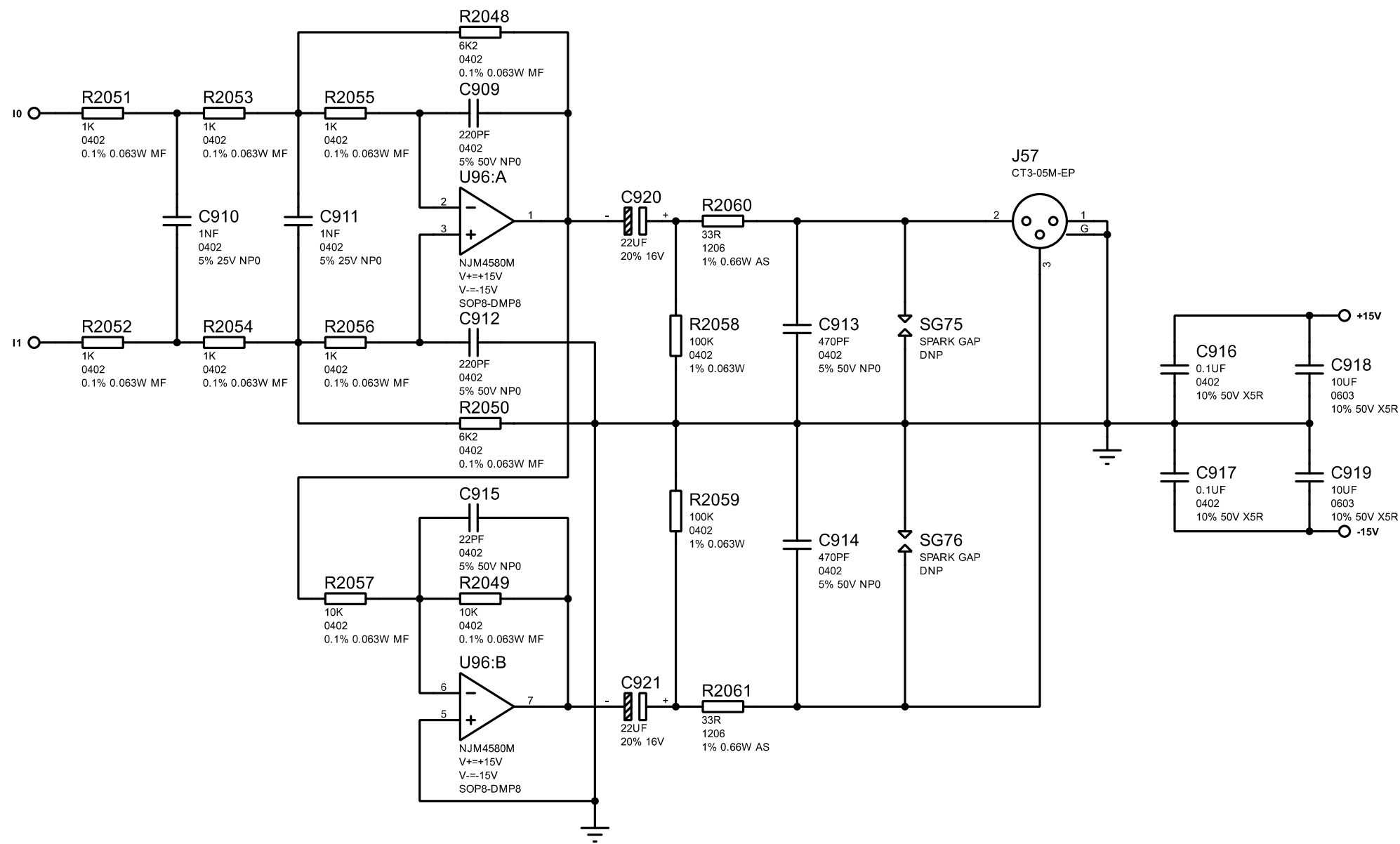
TITLE: D24 Analog		MON LR	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 61/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		SUB_W	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:		Peter Watts	PAGE: 62/64
DATE:		02/03/2026	REV: B



TITLE: D24 Analog		MAIN_L	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:	Peter Watts	PAGE:	63/64
DATE:	02/03/2026	REV:	B



TITLE: D24 Analog		MAIN_R	
FILENAME:		D24 Analog PCBA rev B.pdsprj	
AUTHOR:	Peter Watts	PAGE:	64/64
DATE:	02/03/2026	REV:	B